

DEFENDING THE NETWORK

Fall 2025

Montgomery College

PORTFOLIO

A Collection of Labs & Projects

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Course: NWIT 245 - Defending the Network

Lab Number: 01

Lab Title: Windows 10 Firewall Lab

Professor: Silvia Vargas

Date: 11/09/2025

1. Objective

Enable, configure, and test Windows Defender Firewall on Windows 10.

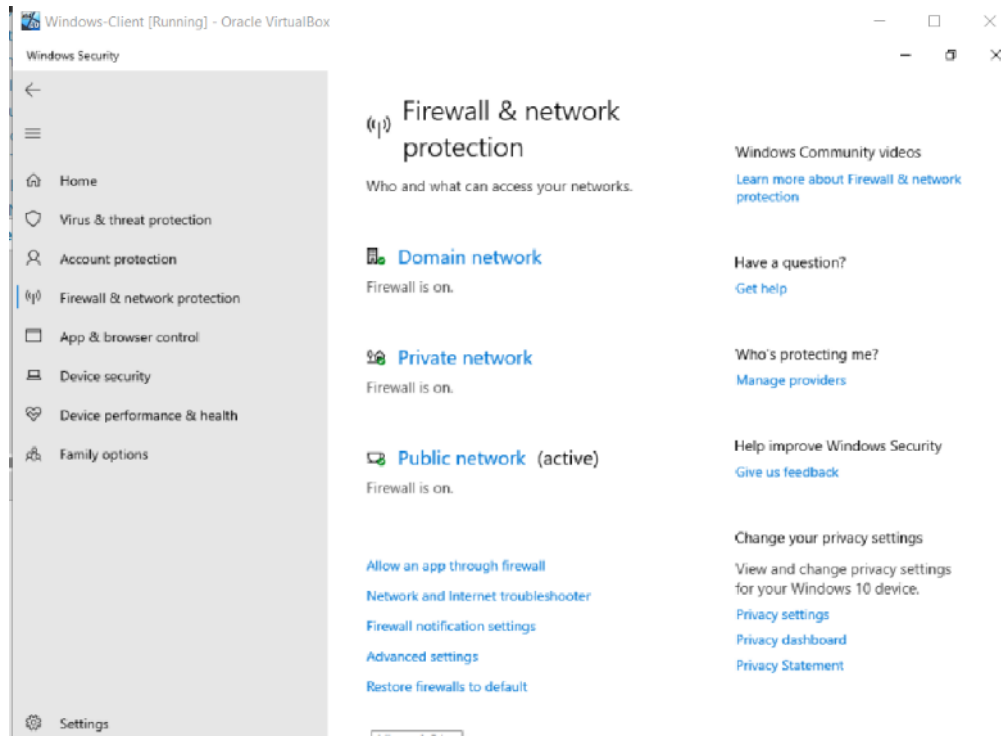
2. Tools and Requirements

- Windows 10 PC or VM with administrator rights
- Basic understanding of ports and network concepts

3. Lab Steps and Execution

Step 1: Verify and Enable Windows Firewall

- Clicked **Start** > typed and opened **Control Panel**
- Went to **System and Security** > **Windows Defender Firewall**
- On the left pane, clicked **Turn Windows Defender Firewall on/off**
- Ensured **Windows Defender Firewall** is turned on for all profiles: Domain, Private, Public



- Opened Command Prompt as Administrator:
- Entered command:

netsh advfirewall show allprofiles

- Verified that State = ON for all profiles

```
Command Prompt
Domain Profile Settings:
-----
State ON
Firewall Policy BlockInbound,AllowOutbound
LocalFirewallRules N/A (GPO-store only)
LocalConSecRules N/A (GPO-store only)
InboundUserNotification Enable
RemoteManagement Disable
UnicastResponseToMulticast Enable

Logging:
LogAllowedConnections Disable
LogDroppedConnections Disable
FileName %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize 4096

Private Profile Settings:
-----
State ON
Firewall Policy BlockInbound,AllowOutbound
LocalFirewallRules N/A (GPO-store only)
LocalConSecRules N/A (GPO-store only)
InboundUserNotification Enable
RemoteManagement Disable
UnicastResponseToMulticast Enable

Logging:
LogAllowedConnections Disable
LogDroppedConnections Disable
```

```
Command Prompt
Public Profile Settings:
-----
State ON
Firewall Policy BlockInbound,AllowOutbound
LocalFirewallRules N/A (GPO-store only)
LocalConSecRules N/A (GPO-store only)
InboundUserNotification Enable
RemoteManagement Disable
UnicastResponseToMulticast Enable

Logging:
LogAllowedConnections Disable
LogDroppedConnections Disable
FileName %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize 4096

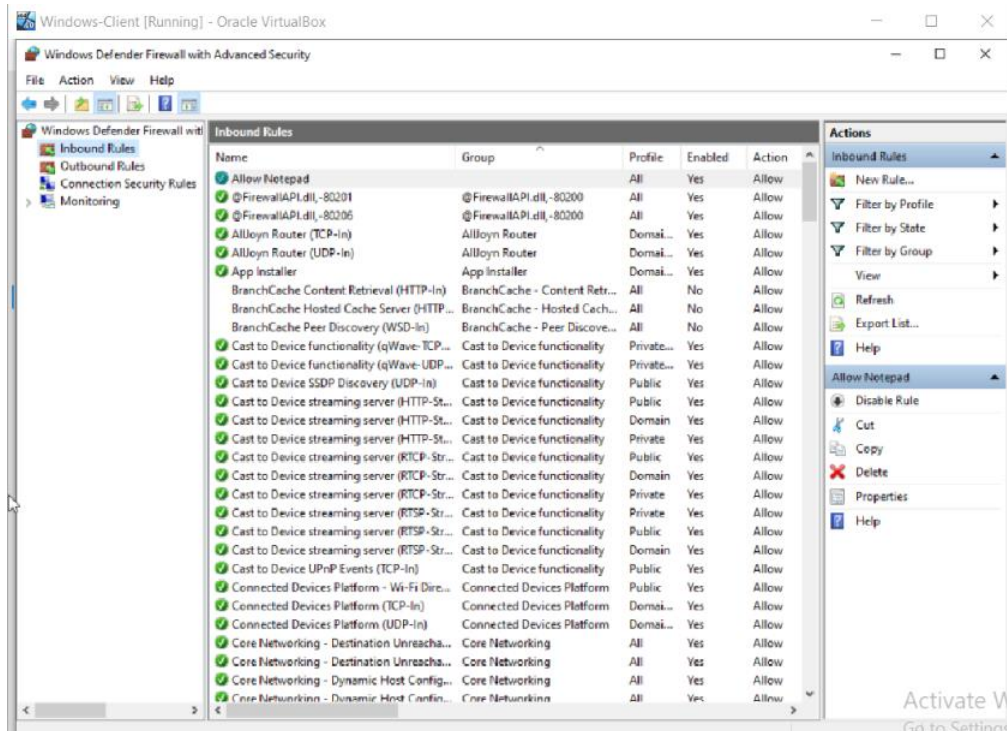
Ok.

C:\Users\win10>
```

Step 2: Create an Inbound Rule to Allow a Program

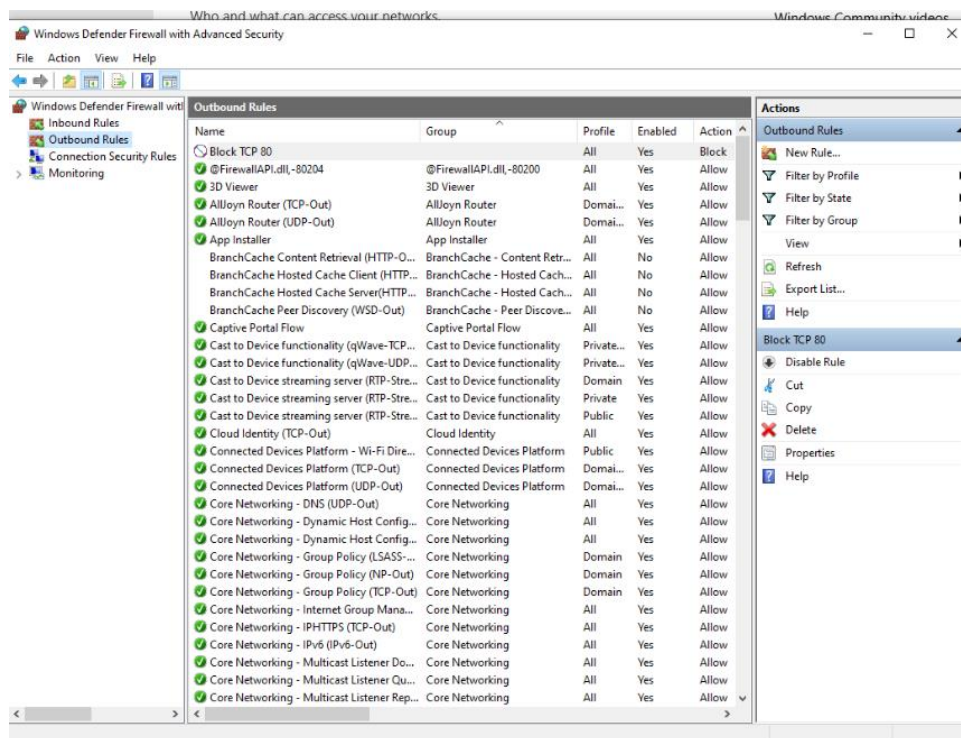
- Opened **Windows Defender Firewall with Advanced Security**:
- Clicked **Advanced settings** (left pane)
- Right-click **Inbound Rules** > **New Rule...**
- Selected **Program** > **Next**
- Clicked **This program path:** > **Browse**
- Navigated to:
 - C:\Windows\System32\notepad.exe
- Selected it > **Open** > **Next**
- Chose **Allow the connection** > **Next**
- Selected all profiles (Domain, Private, Public) > **Next**

- Named the rule "Allow Notepad" > **Finish**



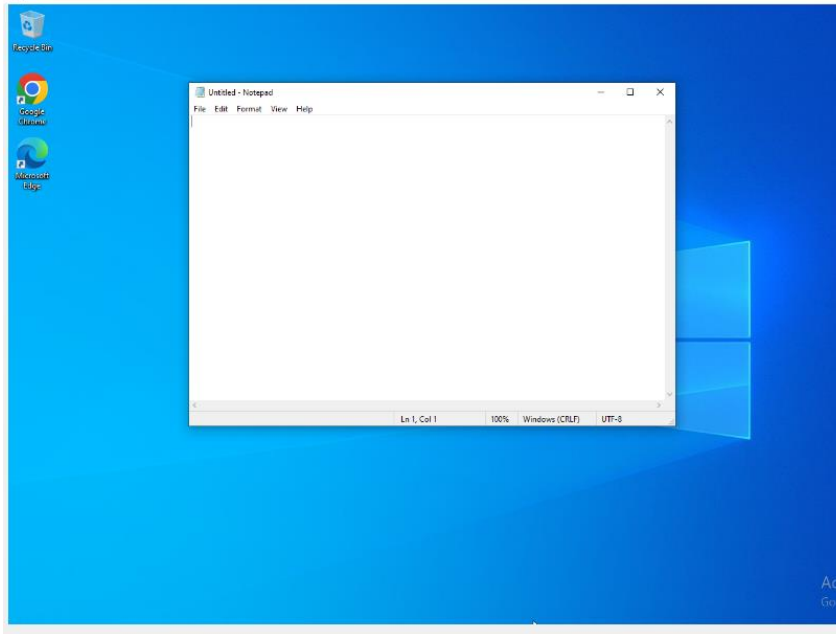
Step 3: Create an Outbound Rule to Block a Port

- In **Windows Defender Firewall with Advanced Security**, right-clicked **Outbound Rules** > **New Rule...**
- Selected **Port** > **Next**
- Selected **TCP** > **Specific local ports: type 80** > **Next**
- Chose **Block the connection** > **Next**
- Selected all profiles (Domain, Private, Public) > **Next**
- Named the rule "Block TCP 80" > **Finish**



Step 4: Test Firewall Rules

- **Test allowed program:**
 - Opened Notepad.

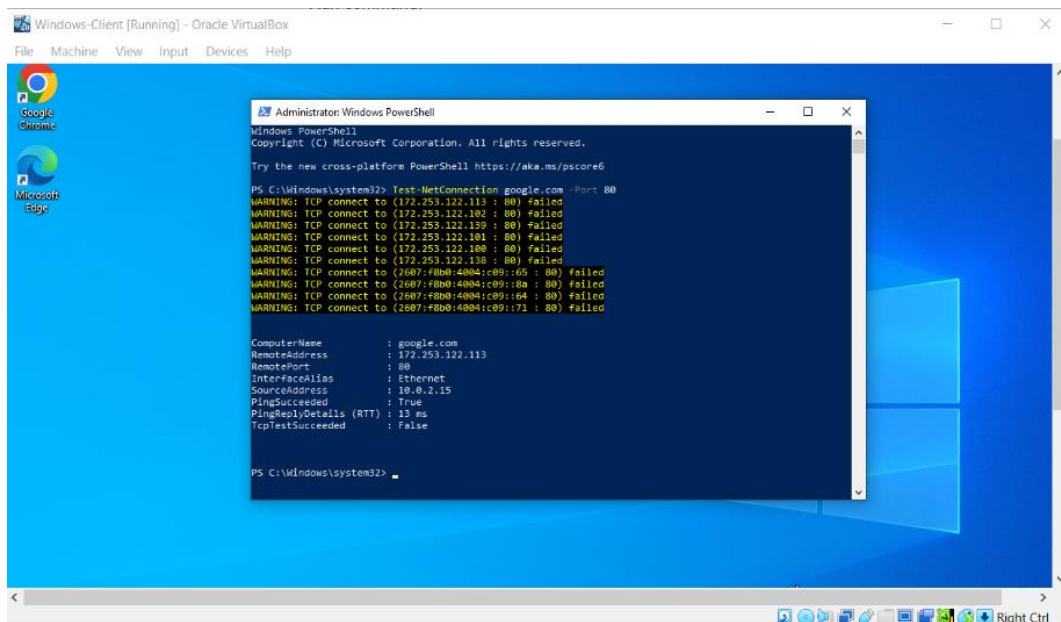


- **Test blocked port:**

- Start > typed powershell > right-clicked **Windows PowerShell** > Run as administrator
- Ran command:

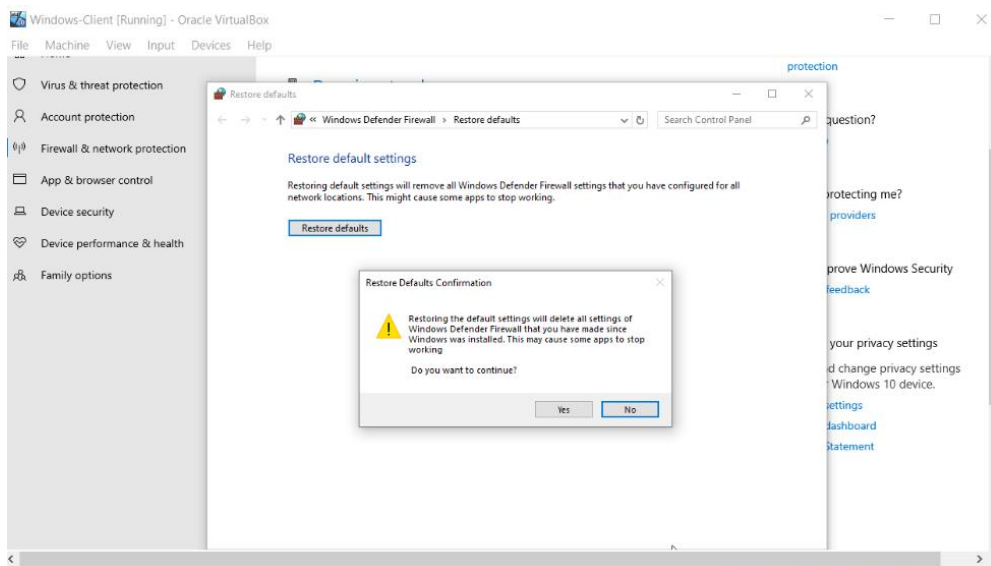
`Test-NetConnection google.com -Port 80`

- Expected result: TcpTestSucceeded : False or connection blocked



Step 5: Reset Firewall to Default Settings

- In **Windows Defender Firewall with Advanced Security**, clicked **Restore Defaults**
- Clicked **Restore defaults** button > **Yes** to confirm



4. Conclusion

This lab successfully enabled, configured, and tested the Windows Defender Firewall. I successfully created an Inbound Rule to allow an application (Notepad) and an Outbound Rule to block a specific TCP port (80). Testing confirmed that the rules functioned as intended, demonstrating practical skills in host-based defense and network traffic control before restoring default settings.

5. Challenges and Observations

- Primary Challenge: The need for administrator rights was essential to perform all configuration and testing steps.
- Key Observation: The lab effectively highlighted the difference between creating firewall rules based on program paths versus those based on specific TCP/UDP ports.
- Verification: Command-line tools like `netsh advfirewall show allprofiles` and `Test-NetConnection` were critical for verifying firewall status and rule efficacy

Course: NWIT 245 - Defending the Network

Lab Number: 02

Lab Title: Kali Linux Firewall Lab

Professor: Silvia Vargas

Date: 11/09/2025

1. Objective

Learn how to enable, configure, and test the firewall using `ufw` (Uncomplicated Firewall) on Kali Linux.

2. Tools and Requirements

- Kali Linux installed (physical or VM)
- Root or sudo privileges
- Basic knowledge of Linux commands and networking

3. Lab Steps and Execution

Step 1: Verify Firewall Status and Enable ufw

- Opened a terminal
- Check the firewall status:
`sudo ufw status verbose`
- Expected: inactive or active
- Enable the firewall if inactive:
`sudo ufw enable`
- Confirmed it is enabled:
`sudo ufw status verbose`

ux-2025.1c-virtualbox-amd64 [Running] - Oracle VirtualBox

hine View Input Devices Help

1 2 3 4

11:34

kali@kali: ~

File Actions Edit View Help

```
(kali@kali)-[~]
$ sudo ufw status verbose
Status: inactive

(kali@kali)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup

(kali@kali)-[~]
$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

(kali@kali)-[~]
$
```

Step 2: Allow an Incoming Program/Service

- Allow SSH traffic (port 22):
sudo ufw allow ssh
- Verify rule added:
sudo ufw status numbered

```
(kali@kali)-[~]
$ sudo ufw allow ssh
Rule added
Rule added (v6)

(kali@kali)-[~]
$ sudo ufw status numbered
Status: active

      To Action From
      --
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 22/tcp (v6) ALLOW IN Anywhere (v6)

(kali@kali)-[~]
$
```

Step 3: Block an Outgoing Port

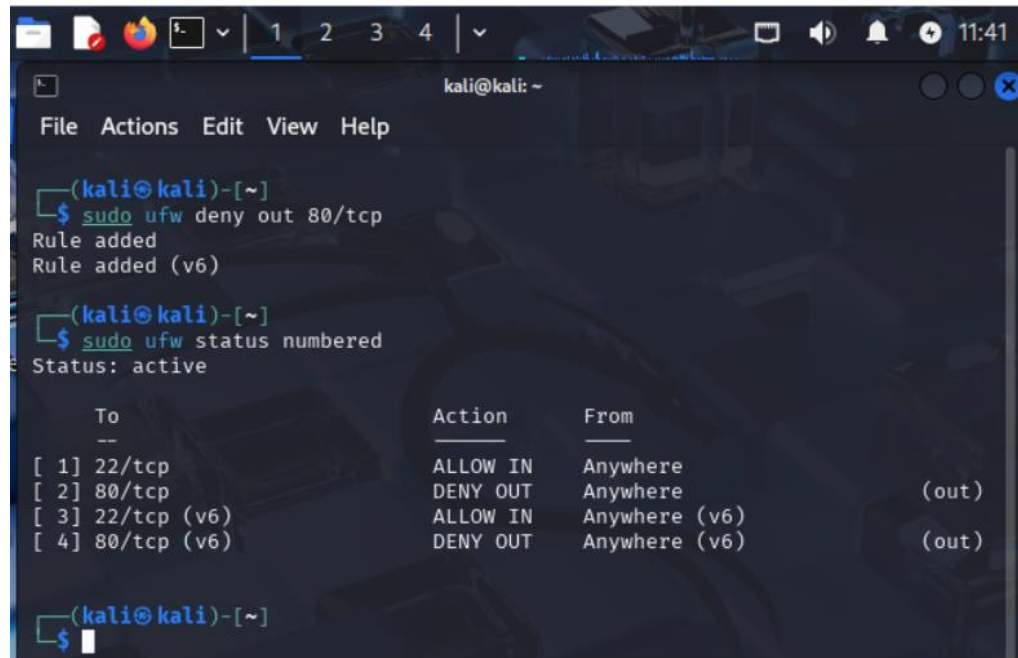
- By default, ufw blocks incoming connections but allows outgoing. To block outgoing port 80:

`sudo ufw deny out 80/tcp`

- Verify rule added:
`sudo ufw status numbered`

ux-2025.1c-virtualbox-amd64 [Running] - Oracle VirtualBox

Machine View Input Devices Help



```

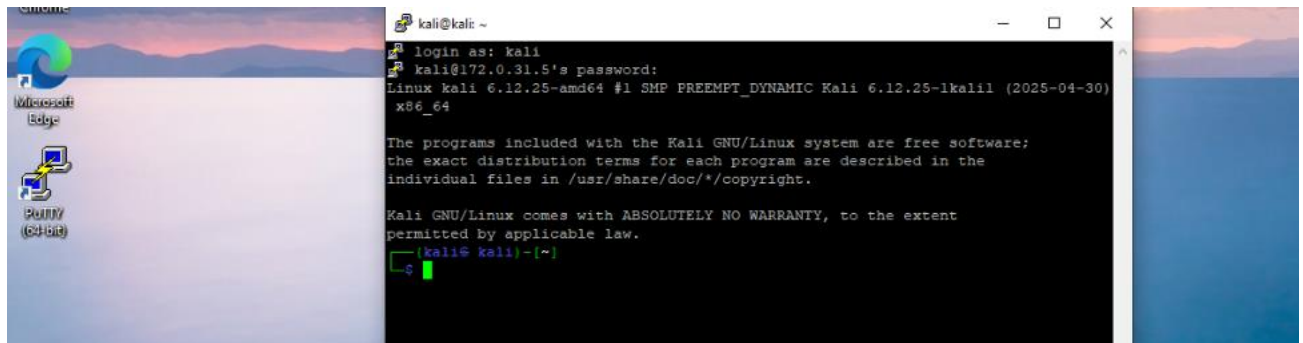
(kali@kali)-[~]
$ sudo ufw deny out 80/tcp
Rule added
Rule added (v6)

(kali@kali)-[~]
$ sudo ufw status numbered
Status: active

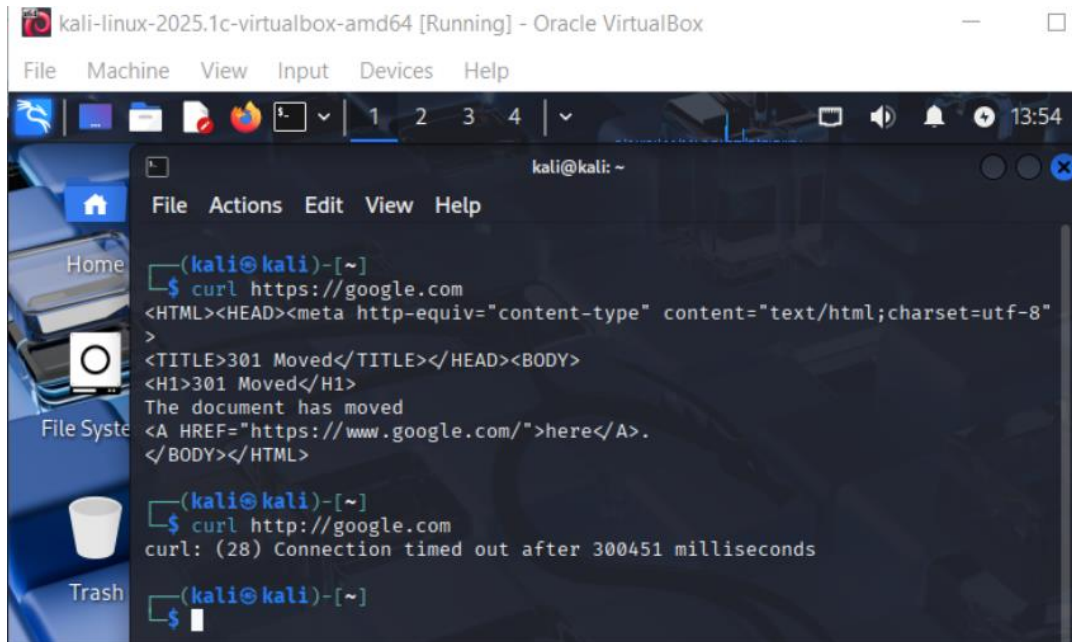
      To Action From
      --
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 80/tcp DENY OUT Anywhere (out)
[ 3] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 4] 80/tcp (v6) DENY OUT Anywhere (v6) (out)
  
```

Step 4: Test the Firewall Rules

- Tested allowed service from a windows terminal
- Successfully SSH into the Kali machine



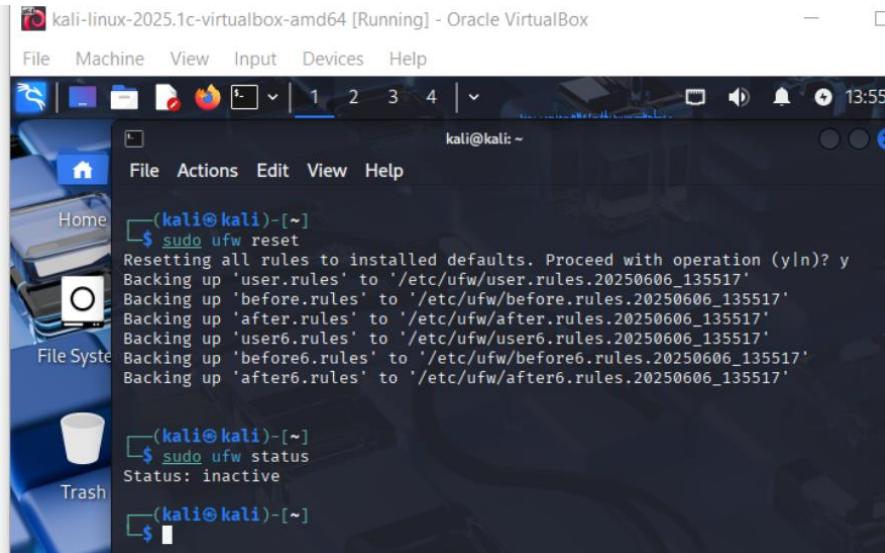
- Tested blocked outgoing port:
- On Kali, run:
`curl http://google.com`
- Connection timed out due to blocked outgoing HTTP.



```
kali-linux-2025.1c-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~
File Actions Edit View Help
Home
File System
Trash
(kali@kali)-[~]
$ curl https://google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="https://www.google.com/">here</A>.
</BODY></HTML>
(kali@kali)-[~]
$ curl http://google.com
curl: (28) Connection timed out after 300451 milliseconds
(kali@kali)-[~]
$
```

Step 5: Reset Firewall Rules

- Reset ufw to default (disables firewall and removes all rules):
`sudo ufw reset`
- Confirm firewall is inactive:
`sudo ufw status`

A screenshot of a Kali Linux terminal window. The window title is 'kali-linux-2025.1c-virtualbox-amd64 [Running] - Oracle VirtualBox'. The terminal shows the following commands and output:

```
(kali@kali)-[~]  
$ sudo ufw reset  
Resetting all rules to installed defaults. Proceed with operation (y/n)? y  
Backing up 'user.rules' to '/etc/ufw/user.rules.20250606_135517'  
Backing up 'before.rules' to '/etc/ufw/before.rules.20250606_135517'  
Backing up 'after.rules' to '/etc/ufw/after.rules.20250606_135517'  
Backing up 'user6.rules' to '/etc/ufw/user6.rules.20250606_135517'  
Backing up 'before6.rules' to '/etc/ufw/before6.rules.20250606_135517'  
Backing up 'after6.rules' to '/etc/ufw/after6.rules.20250606_135517'  
  
(kali@kali)-[~]  
$ sudo ufw status  
Status: inactive  
  
(kali@kali)-[~]  
$
```

Course: NWIT 245 - Defending the Network

Lab Number: 03

Lab Title: Windows 10 and Kali Linux Hardening Lab

Professor: Silvia Vargas

Date: 11/16/2025

1. Objective

Practice basic system hardening techniques on Windows 10 and Kali Linux to improve system security.

2. Tools and Requirements

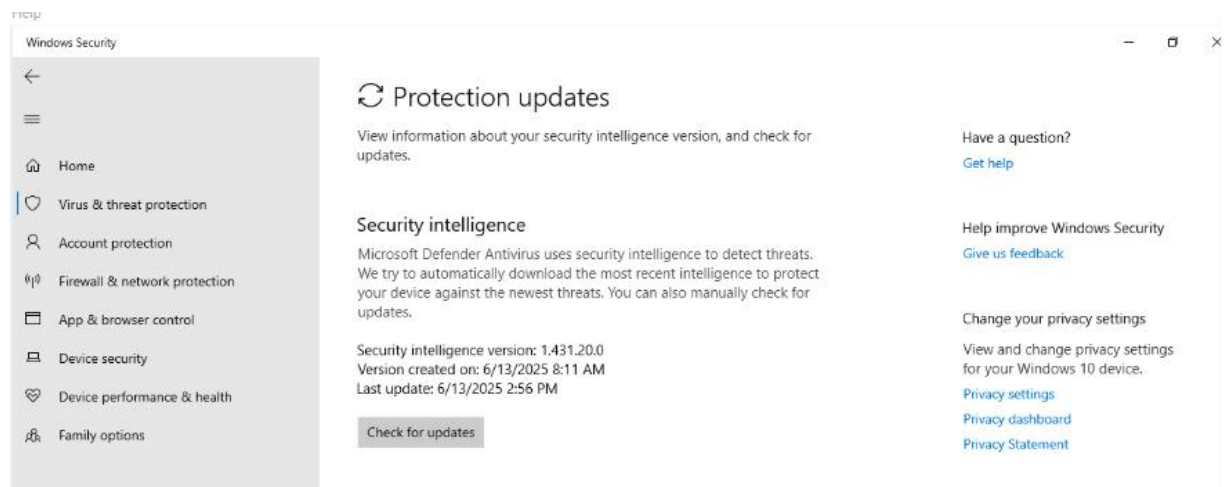
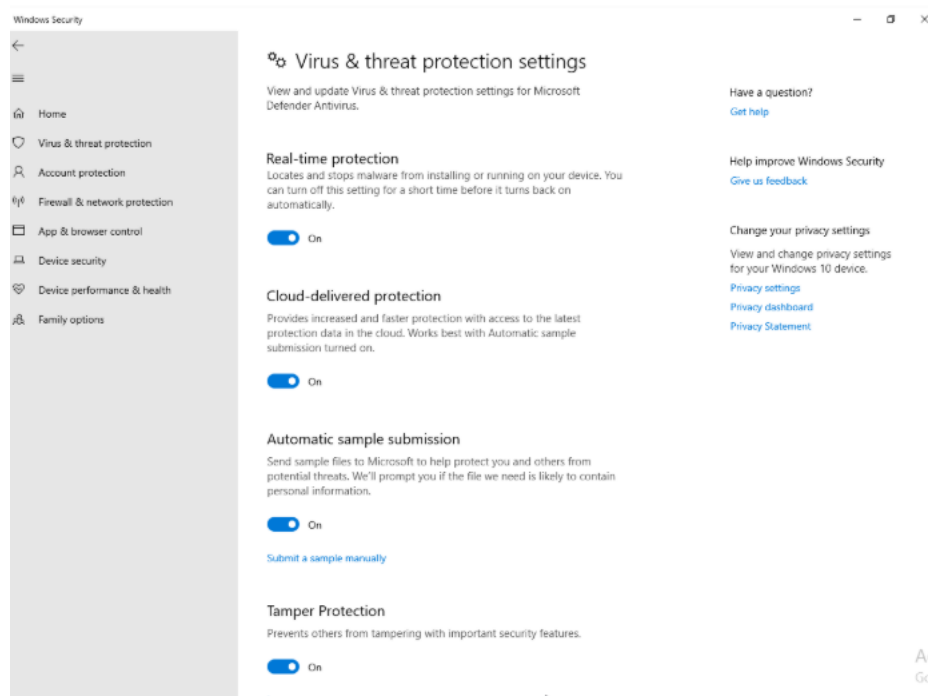
- Windows 10 machine or VM with admin rights
- Kali Linux machine or VM with root or sudo privileges
- Basic understanding of OS security principles

3. Lab Steps and Execution

Part 1: Windows 10 Hardening

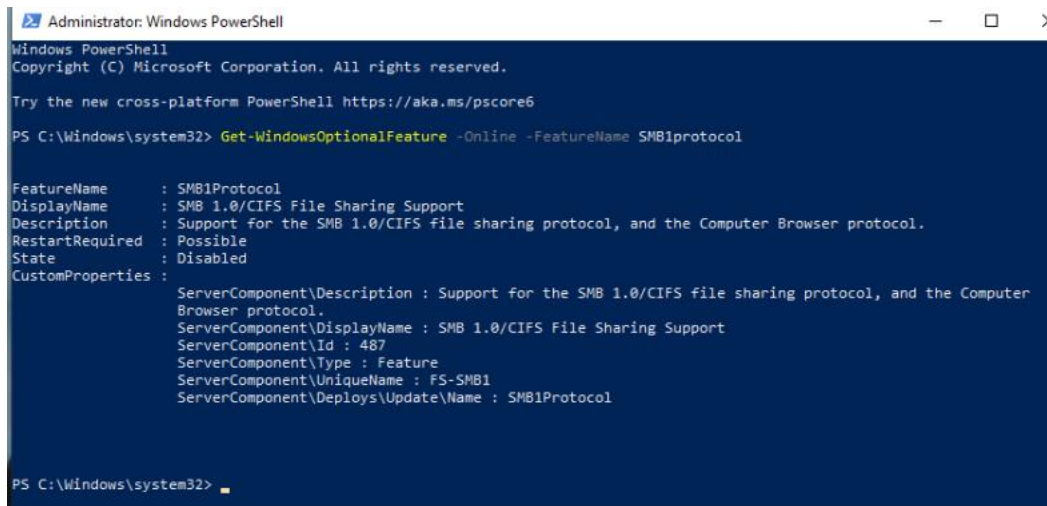
1. Enable Windows Defender Antivirus and Update Definitions

- Open Windows Security from the Start menu
- Click Virus & threat protection
- Ensure Real-time protection is ON
- Click Check for updates under Virus & threat protection updates



2. Disable SMBv1 Protocol

- Open **PowerShell as Administrator**
- Run command:
`Disable-WindowsOptionalFeature -Online -FeatureName smb1protocol`
- Confirm SMBv1 is disabled:
`Get-WindowsOptionalFeature -Online | Where-Object {$_.FeatureName -eq "SMB1Protocol"}`



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

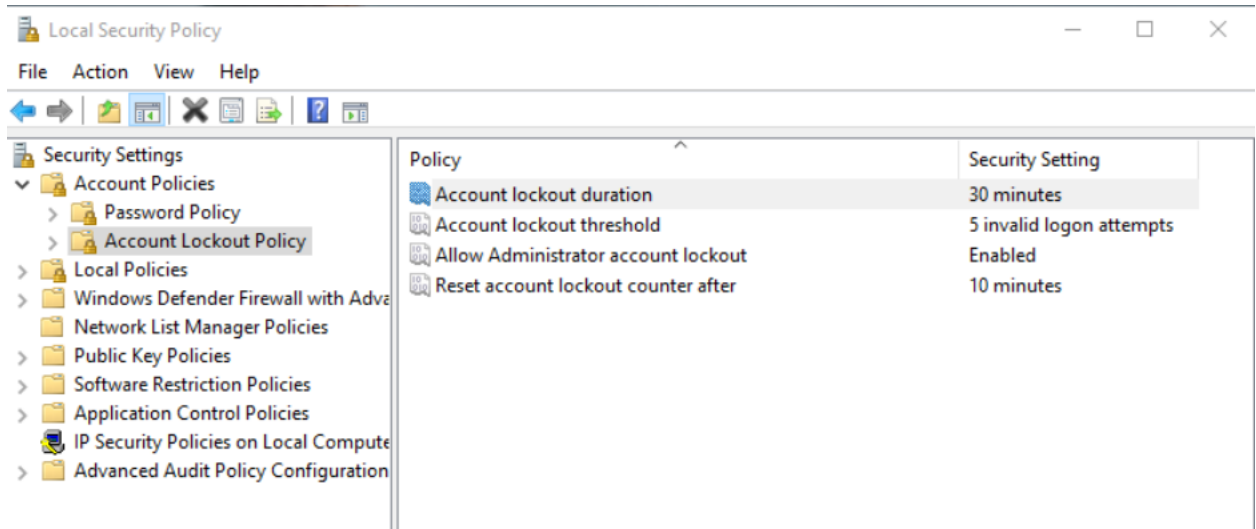
PS C:\Windows\system32> Get-WindowsOptionalFeature -Online -FeatureName SMB1protocol

FeatureName      : SMB1Protocol
DisplayName       : SMB 1.0/CIFS File Sharing Support
Description      : Support for the SMB 1.0/CIFS file sharing protocol, and the Computer Browser protocol.
RestartRequired  : Possible
State            : Disabled
CustomProperties  :
                   ServerComponent\Description : Support for the SMB 1.0/CIFS file sharing protocol, and the Computer
                   Browser protocol.
                   ServerComponent\DisplayName : SMB 1.0/CIFS File Sharing Support
                   ServerComponent\Id          : 487
                   ServerComponent\Type        : Feature
                   ServerComponent\UniqueName  : FS-SMB1
                   ServerComponent\Deploys\Update\Name : SMB1Protocol

PS C:\Windows\system32>
```

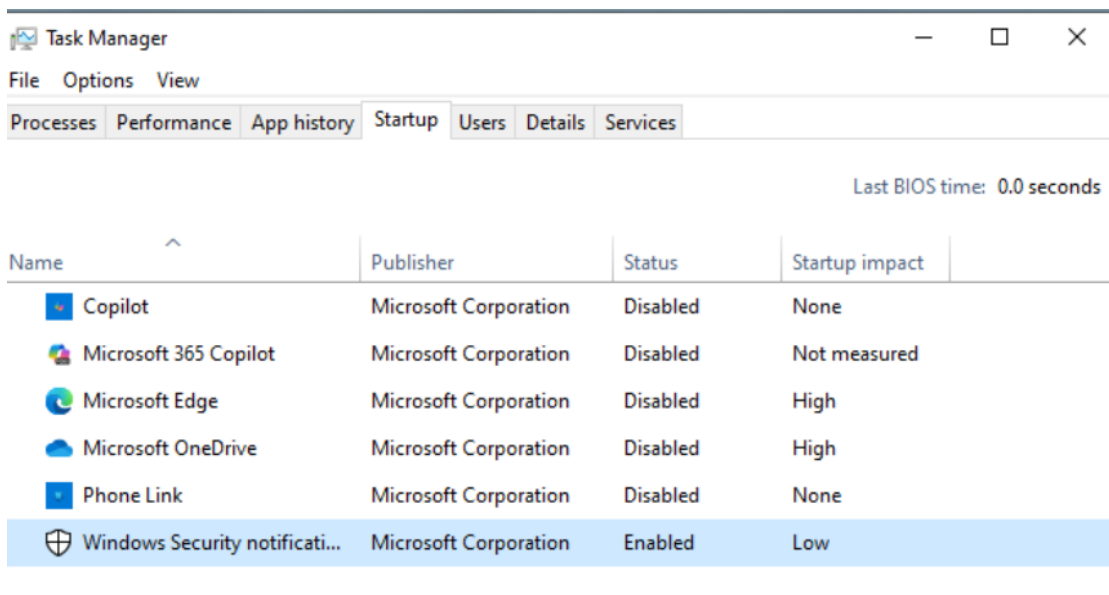
3. Configure Account Lockout Policy

- Opened **Local Security Policy**:
- Start > type secpol.msc > Enter
- Navigated: **Account Policies** > **Account Lockout Policy**
- Set **Account lockout threshold** to 5 invalid attempts
- Set **Account lockout duration** to 30 minutes



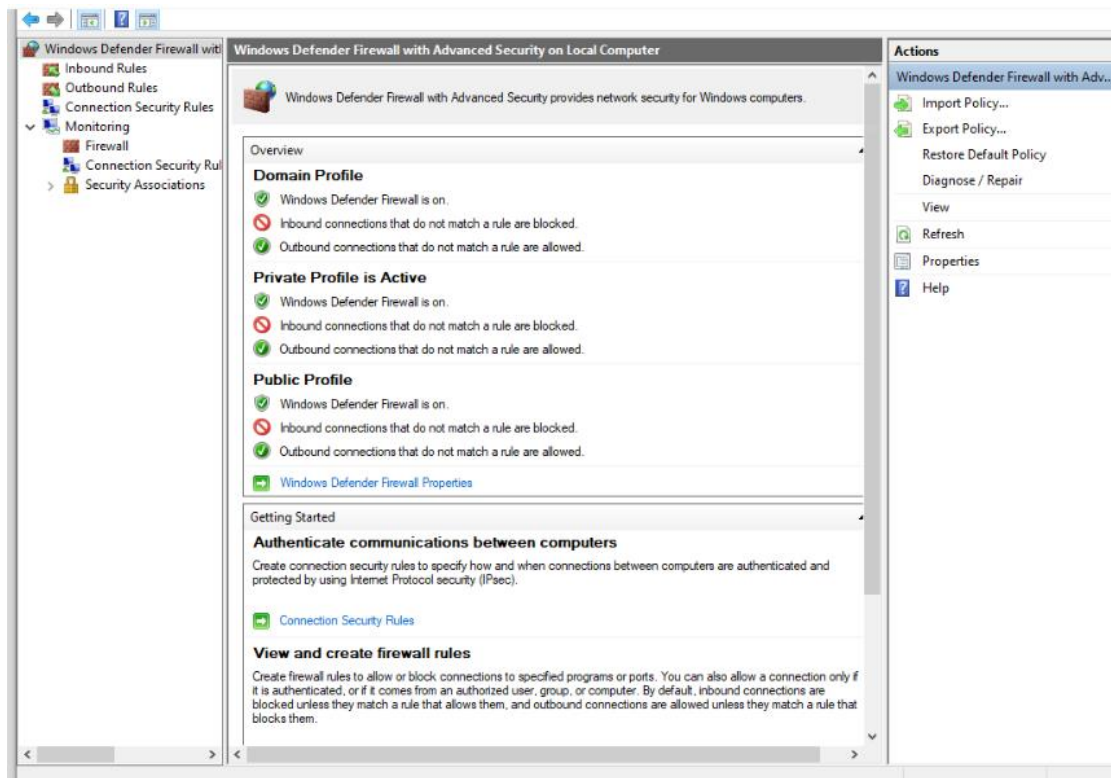
4. Remove Unnecessary Startup Programs

- Opened **Task Manager** (Ctrl + Shift + Esc)
- Went to the **Startup** tab
- Identified non-essential programs and disabled them



5. Enable Windows Firewall with Strict Rules

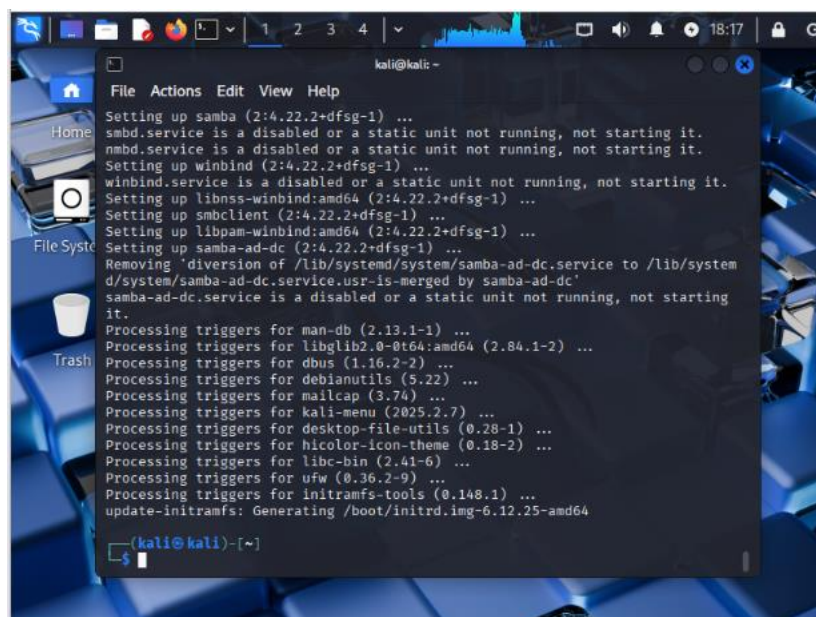
- Opened **Windows Defender Firewall with Advanced Security**
- Verified firewall is ON for all profiles
- Ensured default inbound policy is **Block** and outbound is **Allow**



Part 2: Kali Linux Hardening

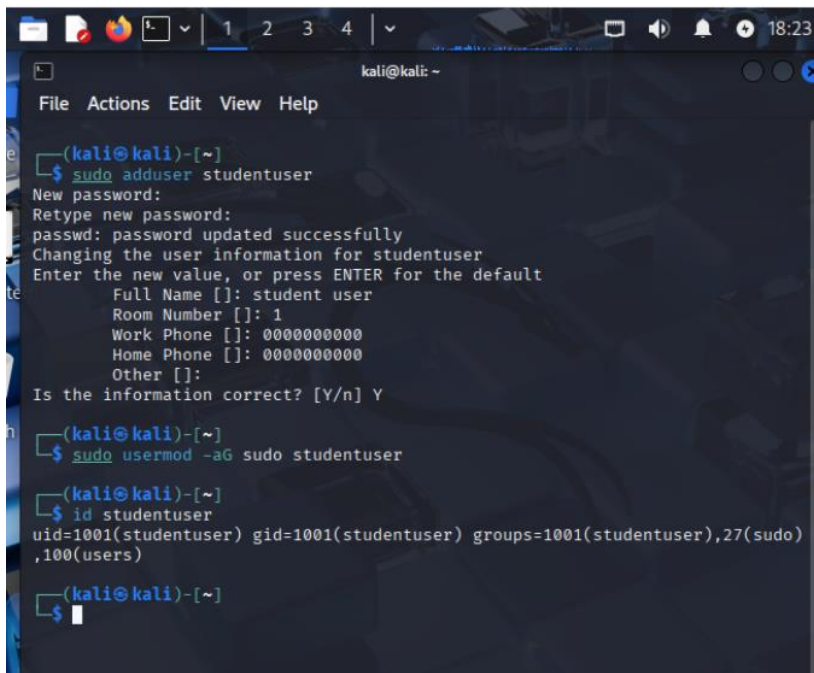
1. Update and Upgrade System Packages

- Opened terminal and run:
sudo apt update && sudo apt upgrade -y



2. Create a New Non-root User and Assign Sudo Privileges

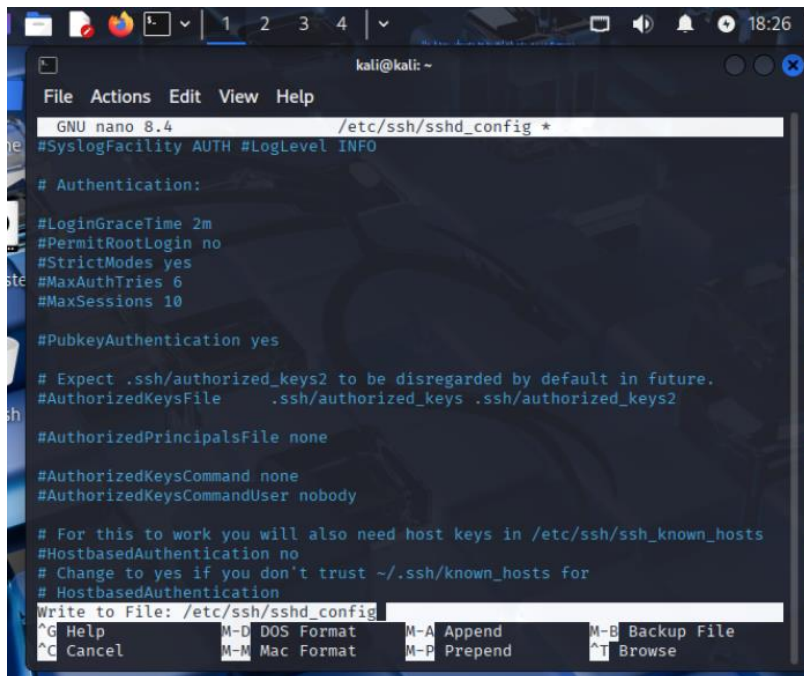
- Run:
`sudo adduser studentuser sudo usermod -aG sudo studentuser`
- Verified user creation:
`id studentuser`



```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ sudo adduser studentuser  
New password:  
Retype new password:  
passwd: password updated successfully  
Changing the user information for studentuser  
Enter the new value, or press ENTER for the default  
Full Name []: student user  
Room Number []: 1  
Work Phone []: 0000000000  
Home Phone []: 0000000000  
Other []:  
Is the information correct? [Y/n] Y  
(kali@kali)-[~]  
$ sudo usermod -aG sudo studentuser  
(kali@kali)-[~]  
$ id studentuser  
uid=1001(studentuser) gid=1001(studentuser) groups=1001(studentuser),27(sudo),100(users)  
(kali@kali)-[~]  
$
```

3. Disable Root SSH Login

- Edit ssh config:
`sudo nano /etc/ssh/sshd_config`
- Find the line `PermitRootLogin` and set it to `no`
- Save and exit, then restart SSH service:
`sudo systemctl restart ssh`

A screenshot of a Kali Linux terminal window. The nano text editor is open, editing the file /etc/ssh/sshd_config. The editor shows various configuration options for SSH, such as #SyslogFacility, #Authentication, #LoginGraceTime, #PermitRootLogin, #StrictModes, #MaxAuthTries, #MaxSessions, #PubkeyAuthentication, #AuthorizedKeysFile, #AuthorizedPrincipalsFile, #AuthorizedKeysCommand, #AuthorizedKeysCommandUser, #HostbasedAuthentication, and #HostbasedAuthentication. The cursor is at the end of the line #HostbasedAuthentication. The bottom of the screen shows the nano editor's command palette with options like Help, Cancel, DOS Format, Mac Format, Append, Prepend, Backup File, and Browse.

```
kali@kali: ~
File Actions Edit View Help
GNU nano 8.4 /etc/ssh/sshd_config *
#SyslogFacility AUTH #LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

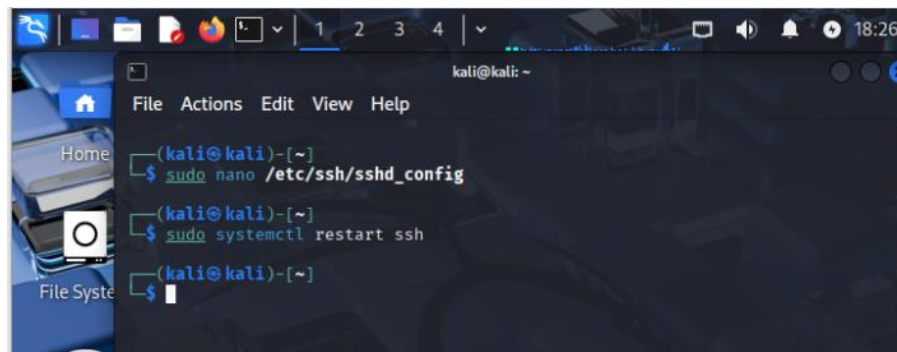
#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

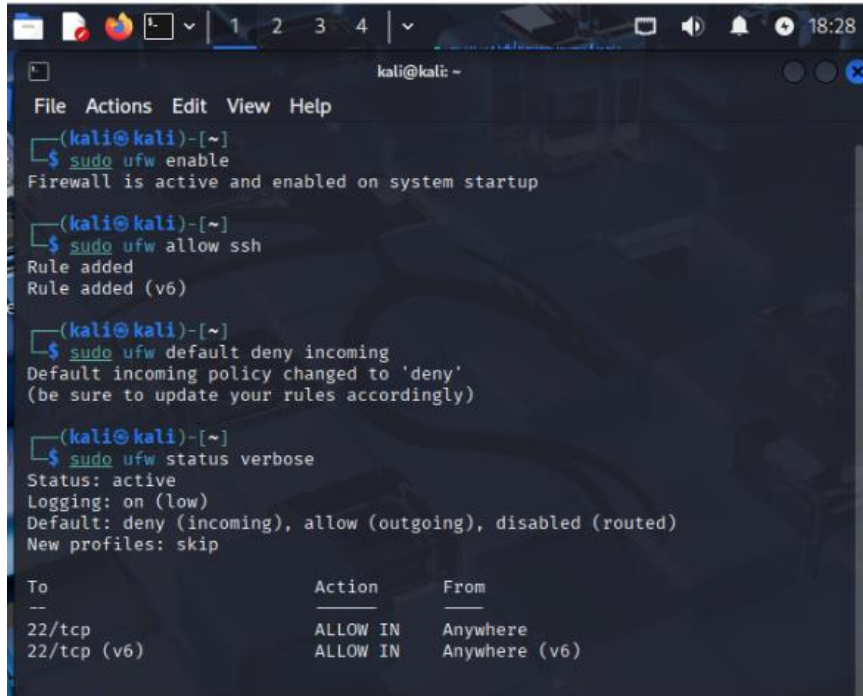
# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
Write to File: /etc/ssh/sshd_config
^G Help      M-D DOS Format  M-A Append     M-B Backup File
^C Cancel    M-M Mac Format  M-P Prepend    ^T Browse
```

A screenshot of a Kali Linux terminal window showing a sequence of commands being executed. The first command is 'sudo nano /etc/ssh/sshd_config', followed by 'sudo systemctl restart ssh'. The terminal shows the prompt '(kali@kali)-[~]' and the command being executed. The second command is 'sudo systemctl restart ssh', and the terminal shows the prompt '(kali@kali)-[~]' and the command being executed. The third command is 'sudo', and the terminal shows the prompt '(kali@kali)-[~]' and the command being executed.

```
kali@kali: ~
File Actions Edit View Help
Home
File System
(kali@kali)-[~]
$ sudo nano /etc/ssh/sshd_config
(kali@kali)-[~]
$ sudo systemctl restart ssh
(kali@kali)-[~]
$ sudo
```

4. Set Up a Basic Firewall with ufw

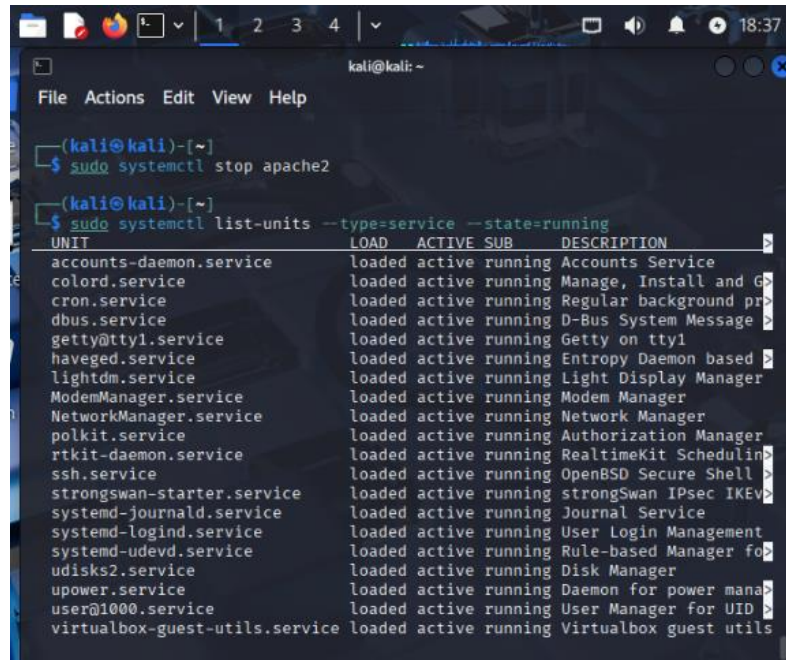
- Enable ufw:
sudo ufw enable
- Allow SSH only:
sudo ufw allow ssh
- Set default incoming policy to deny:
sudo ufw default deny incoming
- Check status:
sudo ufw status verbose



```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ sudo ufw enable  
Firewall is active and enabled on system startup  
  
(kali@kali)-[~]  
$ sudo ufw allow ssh  
Rule added  
Rule added (v6)  
  
(kali@kali)-[~]  
$ sudo ufw default deny incoming  
Default incoming policy changed to 'deny'  
(be sure to update your rules accordingly)  
  
(kali@kali)-[~]  
$ sudo ufw status verbose  
Status: active  
Logging: on (low)  
Default: deny (incoming), allow (outgoing), disabled (routed)  
New profiles: skip  
  
To Action From  
--  
22/tcp ALLOW IN Anywhere  
22/tcp (v6) ALLOW IN Anywhere (v6)
```

5. Remove Unused Services

- List running services:
 `sudo systemctl list-units --type=service --state=running`
- Disable unnecessary services, e.g.:
 `sudo systemctl disable apache2`



```
(kali@kali)-[~]
$ sudo systemctl stop apache2

(kali@kali)-[~]
$ sudo systemctl list-units --type=service --state=running
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
accounts-daemon.service	loaded	active	running	Accounts Service
colord.service	loaded	active	running	Manage, Install and G>
cron.service	loaded	active	running	Regular background pr>
dbus.service	loaded	active	running	D-Bus System Message >
getty@tty1.service	loaded	active	running	Getty on tty1
haveged.service	loaded	active	running	Entropy Daemon based >
lightdm.service	loaded	active	running	Light Display Manager
ModemManager.service	loaded	active	running	Modem Manager
NetworkManager.service	loaded	active	running	Network Manager
polkit.service	loaded	active	running	Authorization Manager
rtkit-daemon.service	loaded	active	running	RealtimeKit Schedulin>
ssh.service	loaded	active	running	OpenBSD Secure Shell >
strongswan-starter.service	loaded	active	running	strongSwan IPsec IKEv>
systemd-journald.service	loaded	active	running	Journal Service
systemd-logind.service	loaded	active	running	User Login Management
systemd-udevd.service	loaded	active	running	Rule-based Manager fo>
udisks2.service	loaded	active	running	Disk Manager
upower.service	loaded	active	running	Daemon for power mana>
user@1000.service	loaded	active	running	User Manager for UID >
virtualbox-guest-utils.service	loaded	active	running	Virtualbox guest utils

Course: NWIT 245 - Defending the Network

Lab Number: 04

Lab Title: Windows 10 and Kali Linux Patching & Vulnerability Assessment Lab

Professor: Silvia Vargas

Date: 11/23/2025

1. Objective

Learn how to apply patches, run vulnerability scans, and assess system security on Windows 10 and Kali Linux.

2. Tools and Requirements

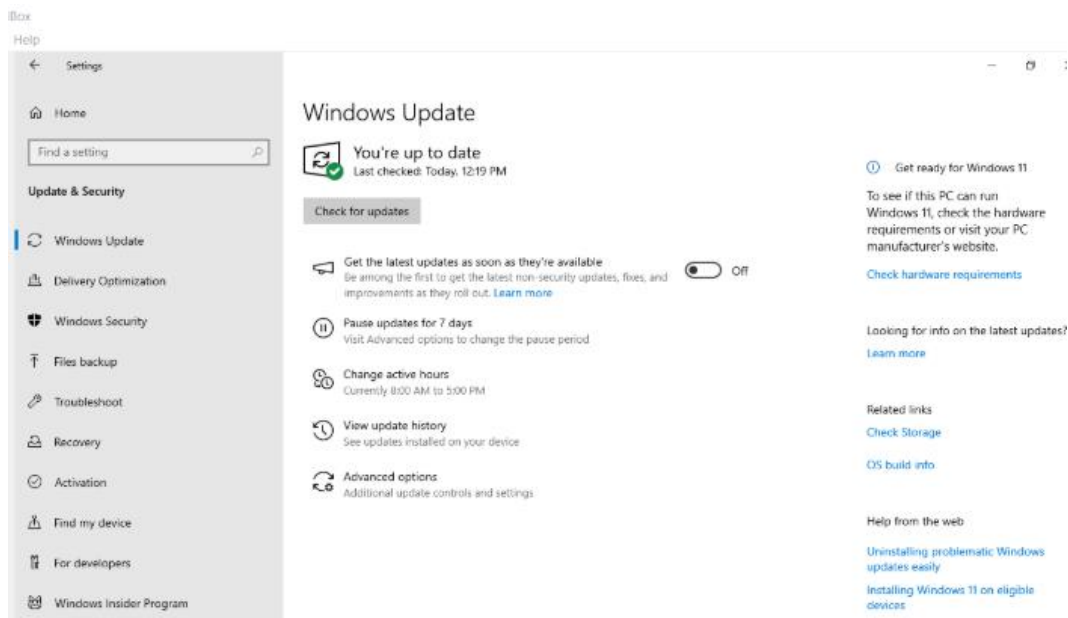
- Windows 10 machine or VM with administrator privileges
- Kali Linux machine or VM with root or sudo access
- Basic knowledge of patch management and vulnerability scanning

3. Lab Steps and Execution

Part 1: Windows 10 Patching and Vulnerability Assessment

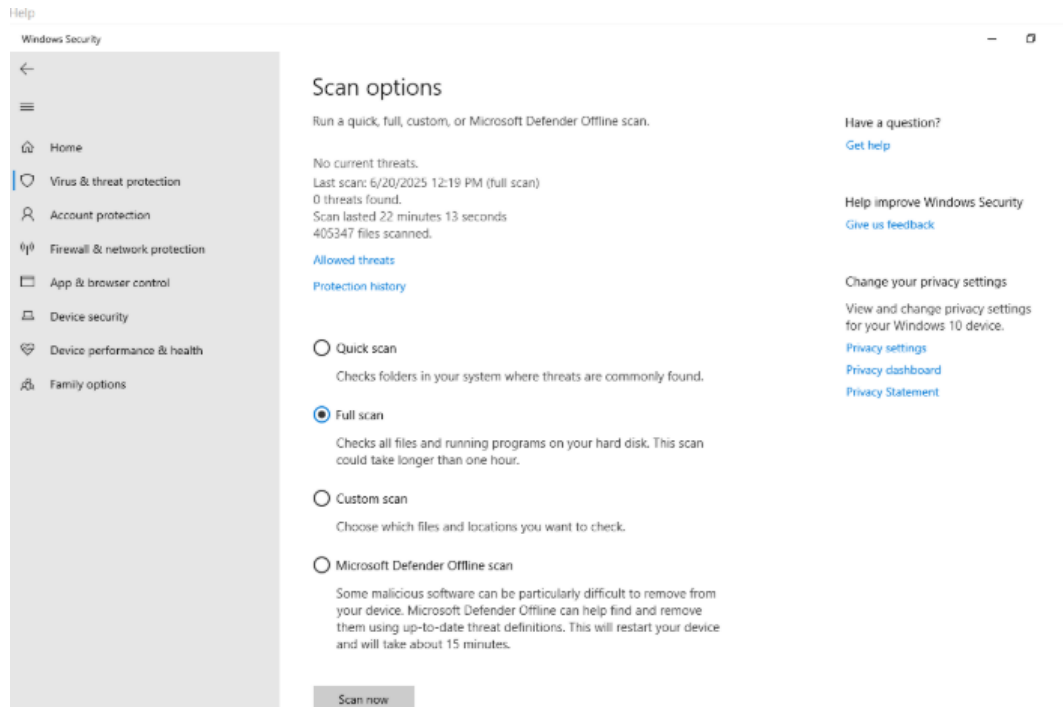
1. Check for Windows Updates and Install Patches

- Open **Settings > Update & Security > Windows Update**
- Click **Check for updates**
- Checked for updates and the system is currently up to date



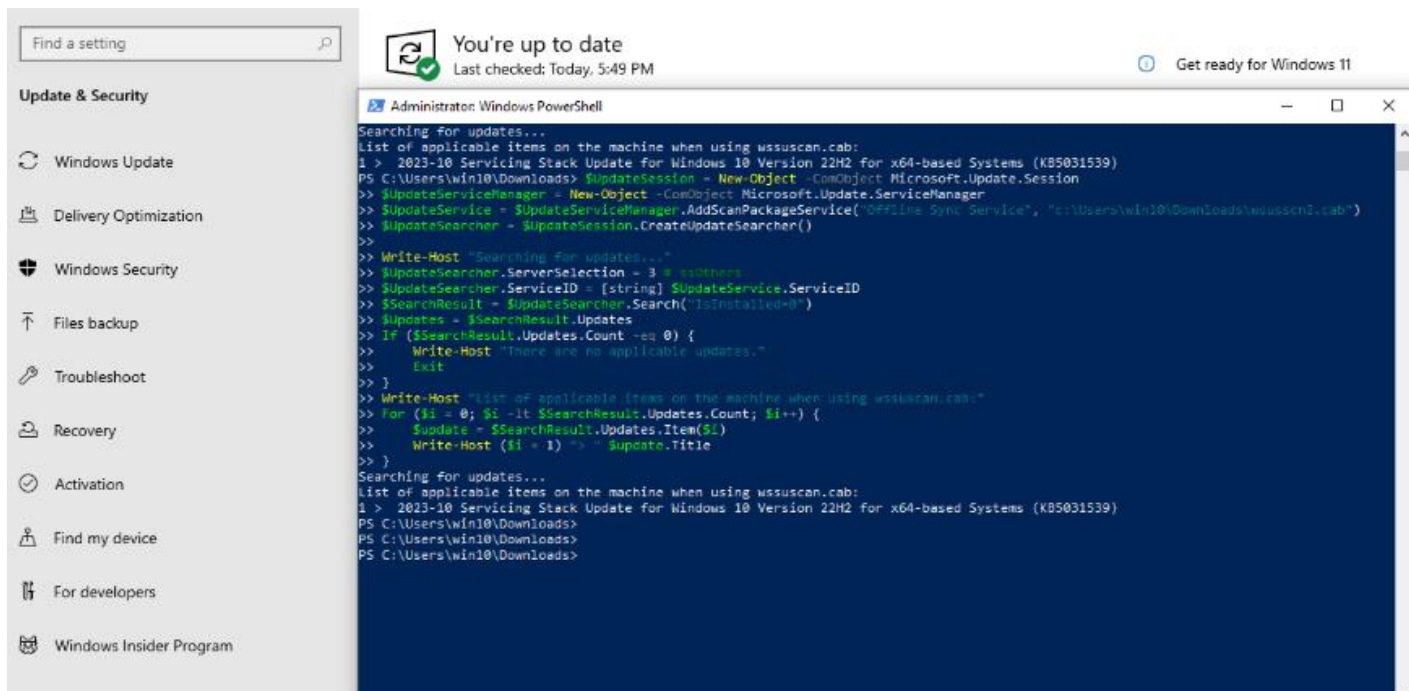
2. Run Windows Defender Antivirus Full Scan

- Open **Windows Security > Virus & threat protection**
- Click **Scan options > select Full scan > Scan now**
- Analyzed scan, no threats were found.



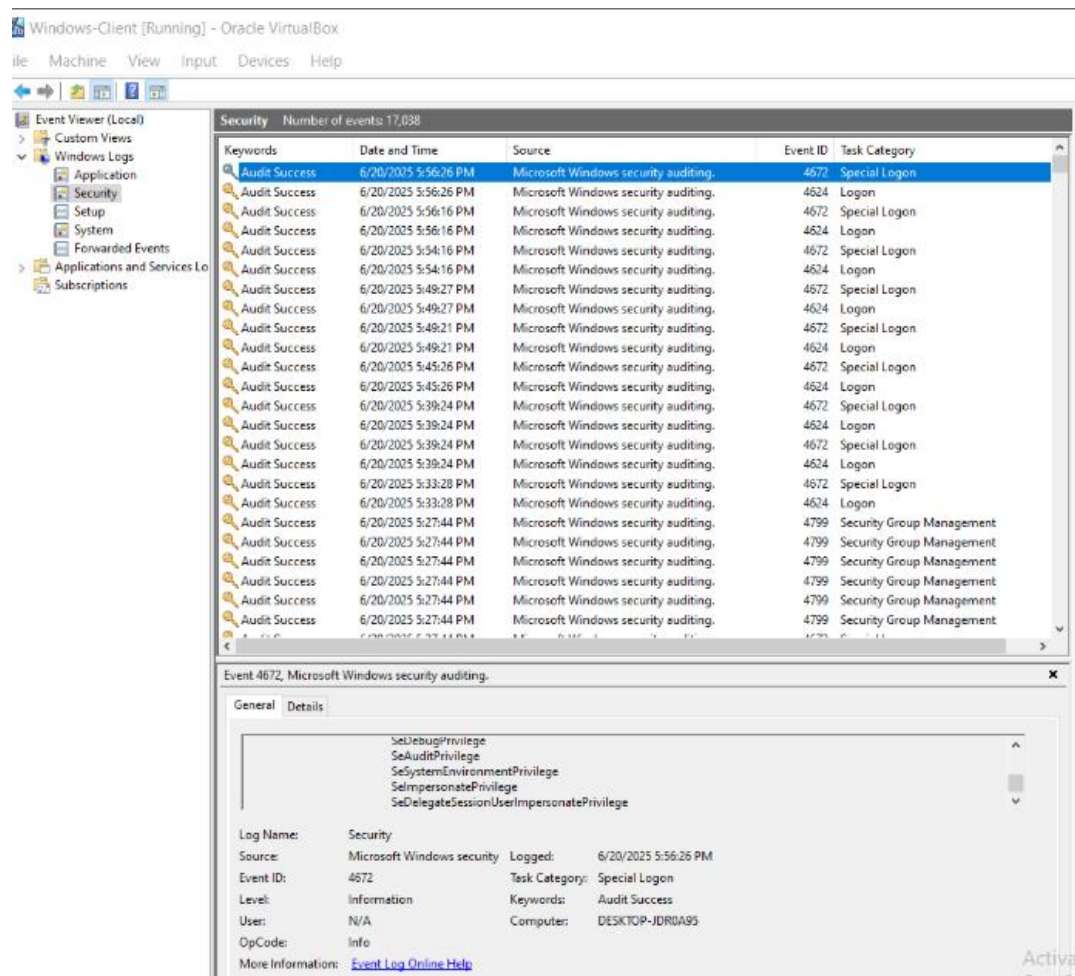
3. Download and Run Microsoft Baseline Security Analyzer (MBSA)

- Download MBSA tool from Microsoft website
- Install and run MBSA
- Scan the local system for missing patches and vulnerabilities
- An update from 10/2023 is available to install.



4. Review Event Viewer for Security Logs

- Open **Event Viewer** (Start > type eventvwr.msc)
- Navigate to **Windows Logs > Security**
- Identify any recent security warnings or errors

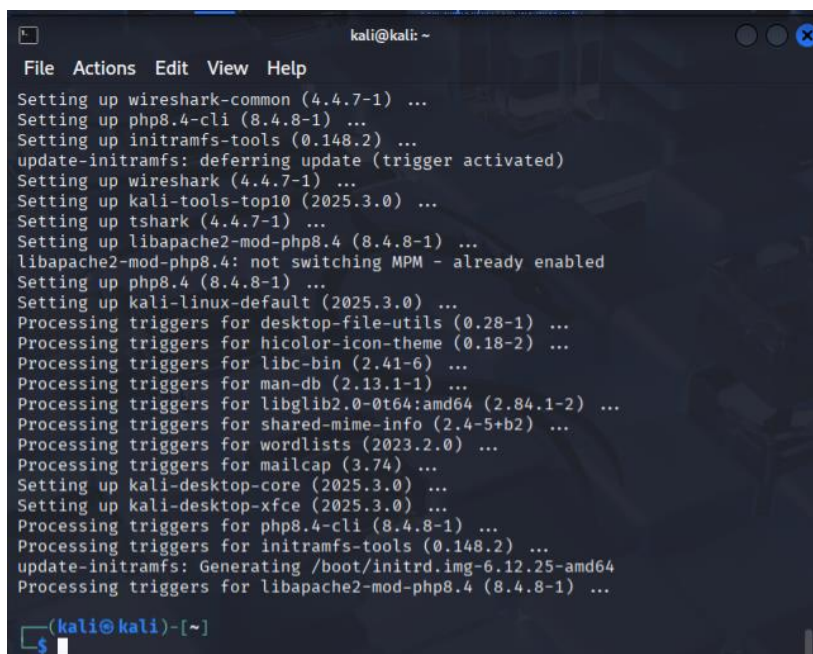


Part 2: Kali Linux Patching and Vulnerability Assessment

1. Update and Upgrade Kali Linux Packages

- Open terminal and run:

```
sudo apt update && sudo apt upgrade -y
```
- Confirm no errors and all packages are upgraded



```
kali@kali: ~
File Actions Edit View Help
Setting up wireshark-common (4.4.7-1) ...
Setting up php8.4-cli (8.4.8-1) ...
Setting up initramfs-tools (0.148.2) ...
update-initramfs: deferring update (trigger activated)
Setting up wireshark (4.4.7-1) ...
Setting up kali-tools-top10 (2025.3.0) ...
Setting up tshark (4.4.7-1) ...
Setting up libapache2-mod-php8.4 (8.4.8-1) ...
libapache2-mod-php8.4: not switching MPM - already enabled
Setting up php8.4 (8.4.8-1) ...
Setting up kali-linux-default (2025.3.0) ...
Processing triggers for desktop-file-utils (0.28-1) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for libc-bin (2.41-6) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libglib2.0-0t64:amd64 (2.84.1-2) ...
Processing triggers for shared-mime-info (2.4-5+b2) ...
Processing triggers for wordlists (2023.2.0) ...
Processing triggers for mailcap (3.74) ...
Setting up kali-desktop-core (2025.3.0) ...
Setting up kali-desktop-xfce (2025.3.0) ...
Processing triggers for php8.4-cli (8.4.8-1) ...
Processing triggers for initramfs-tools (0.148.2) ...
update-initramfs: Generating /boot/initrd.img-6.12.25-amd64
Processing triggers for libapache2-mod-php8.4 (8.4.8-1) ...
(kali@kali)-[~]
$
```

2. Install and Run lynis Security Audit Tool

- Install lynis:
sudo apt install lynis -y
- Run a full system audit:
sudo lynis audit system

```
kali@kali: ~  
File Actions Edit View Help  
  
Lynis security scan details:  
  
Hardening index : 63 [#####] ]  
Tests performed : 272  
Plugins enabled : 1  
  
Components:  
- Firewall [V]  
- Malware scanner [X]  
  
Scan mode:  
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]  
  
Lynis modules:  
- Compliance status [?]  
- Security audit [V]  
- Vulnerability scan [V]  
  
Files:  
- Test and debug information : /var/log/lynis.log  
- Report data : /var/log/lynis-report.dat  
  
Lynis 3.1.4
```

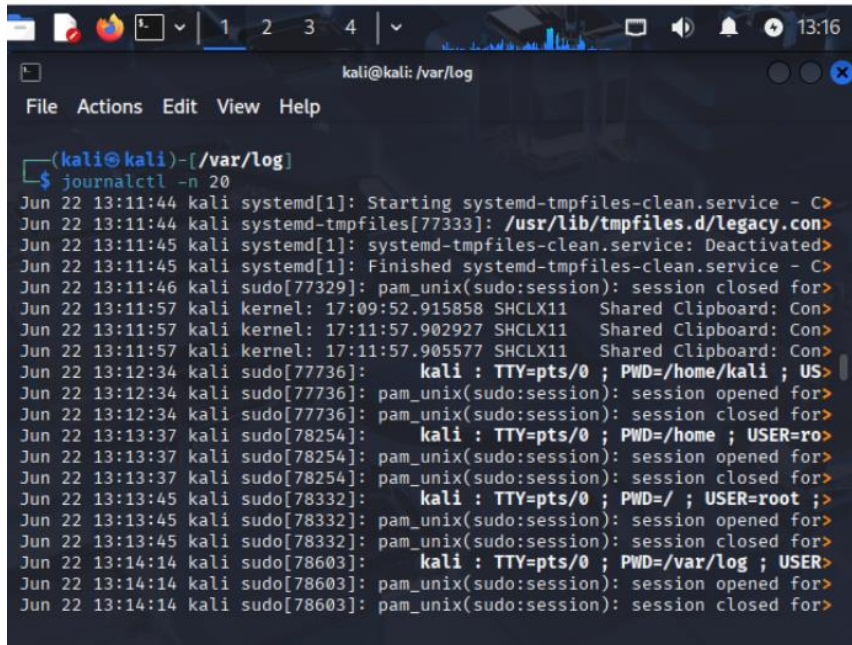
3. Use nmap for Local Vulnerability Scan

- Run a scan of the local system:
 sudo nmap -sV -O localhost
- Note open ports, services, and detected OS

```
kali@kali: ~  
File Actions Edit View Help  
  
2007-2024, CISOfy - https://cisofy.com/lynis/  
Enterprise support available (compliance, plugins, interface and tools)  
  
[TIP]: Enhance Lynis audits by adding your settings to custom.prfl (see /etc/  
/lynis/default.prfl for all settings)  
  
(kali@kali)-[~]  
$ sudo nmap -sV -O localhost  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-06-22 13:11 EDT  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000022s latency).  
Other addresses for localhost (not scanned): ::1  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 0 hops  
  
OS and Service detection performed. Please report any incorrect results at ht  
tps://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.76 seconds  
  
(kali@kali)-[~]  
$
```

4. Review System Logs for Security Events

- Check recent syslog messages related to security:
`sudo tail -n 20 /var/log/auth.log`
- Look for any failed login attempts or errors



```
(kali@kali)-[/var/log]
$ journalctl -n 20
Jun 22 13:11:44 kali systemd[1]: Starting systemd-tmpfiles-clean.service - C>
Jun 22 13:11:44 kali systemd-tmpfiles[77333]: /usr/lib/tmpfiles.d/legacy.con>
Jun 22 13:11:45 kali systemd[1]: systemd-tmpfiles-clean.service: Deactivated>
Jun 22 13:11:45 kali systemd[1]: Finished systemd-tmpfiles-clean.service - C>
Jun 22 13:11:46 kali sudo[77329]: pam_unix(sudo:session): session closed for>
Jun 22 13:11:57 kali kernel: 17:09:52.915858 SHCLX11 Shared Clipboard: Con>
Jun 22 13:11:57 kali kernel: 17:11:57.902927 SHCLX11 Shared Clipboard: Con>
Jun 22 13:11:57 kali kernel: 17:11:57.905577 SHCLX11 Shared Clipboard: Con>
Jun 22 13:12:34 kali sudo[77736]:      kali : TTY=pts/0 ; PWD=/home/kali ; US>
Jun 22 13:12:34 kali sudo[77736]: pam_unix(sudo:session): session opened for>
Jun 22 13:12:34 kali sudo[77736]: pam_unix(sudo:session): session closed for>
Jun 22 13:13:37 kali sudo[78254]:      kali : TTY=pts/0 ; PWD=/home ; USER=ro>
Jun 22 13:13:37 kali sudo[78254]: pam_unix(sudo:session): session opened for>
Jun 22 13:13:37 kali sudo[78254]: pam_unix(sudo:session): session closed for>
Jun 22 13:13:45 kali sudo[78332]:      kali : TTY=pts/0 ; PWD=/ ; USER=root ;>
Jun 22 13:13:45 kali sudo[78332]: pam_unix(sudo:session): session opened for>
Jun 22 13:13:45 kali sudo[78332]: pam_unix(sudo:session): session closed for>
Jun 22 13:14:14 kali sudo[78603]:      kali : TTY=pts/0 ; PWD=/var/log ; USER>
Jun 22 13:14:14 kali sudo[78603]: pam_unix(sudo:session): session opened for>
Jun 22 13:14:14 kali sudo[78603]: pam_unix(sudo:session): session closed for>
```

Course: NWIT 245 - Defending the Network

Lab Number: 05

Lab Title: NIST SP 800-53 Security Controls Lab

Professor: Silvia Vargas

Date: 12/07/2025

1. Objective

Understand and apply select NIST SP 800-53 security controls on a Windows 10 and Kali Linux system to improve organizational security posture.

2. Tools and Requirements

- Windows 10 machine or VM with admin rights
- Kali Linux machine or VM with root or sudo privileges
- Basic understanding of cybersecurity frameworks and security controls

3. Lab Steps and Execution

Step 1: Identify Applicable Controls (Select 3 Controls)

- Review the NIST SP 800-53 Revision 5 catalog or a summarized list
- Select three (3) security controls relevant to endpoint hardening or system security (e.g., AC-2 Account Management, SI-3 Malicious Code Protection, AU-6 Audit Review)
- Document your chosen controls with a brief description of each and why they apply to your environment

Security controls:

CM-02 Baseline Configuration - Baseline configurations document agreed upon system specifications, covering connectivity, operations, and communication, serving as a foundation for future changes. They establish security settings and specifications for devices by documenting these configurations. Organizations can ensure consistent implementation of security controls, such as antivirus software, firewall settings, and access controls across all endpoints.

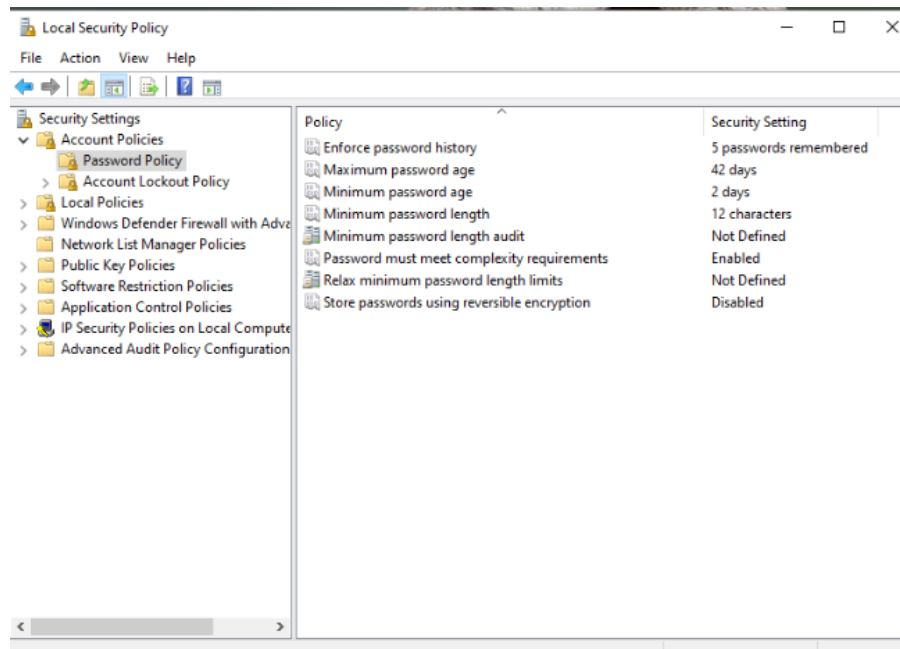
AC-18 Wireless Access - Wireless networks use authentication protocols that provide authenticator protection and mutual authentication. Maintaining strict controls and policies for device connectivity helps ensure that only compliant and secured devices can access the network, thereby enhancing overall endpoint security.

RA-05 Vulnerability Monitoring & Configuration - Security categorization of information and systems informs the frequency and thoroughness of vulnerability monitoring, ensuring that potential sources of vulnerabilities like infrastructure

components and networked devices are effectively scanned and addressed. The principles of vulnerability monitoring are essential for enhancing endpoint security, as they ensure consistent scanning and assessment of devices connected to the network

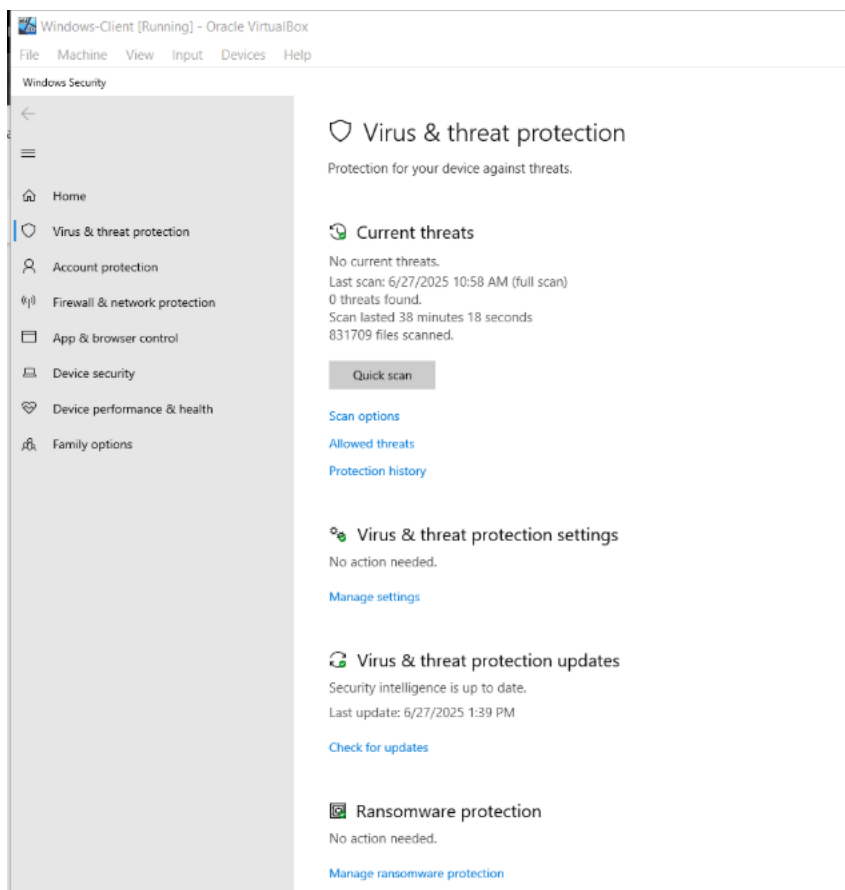
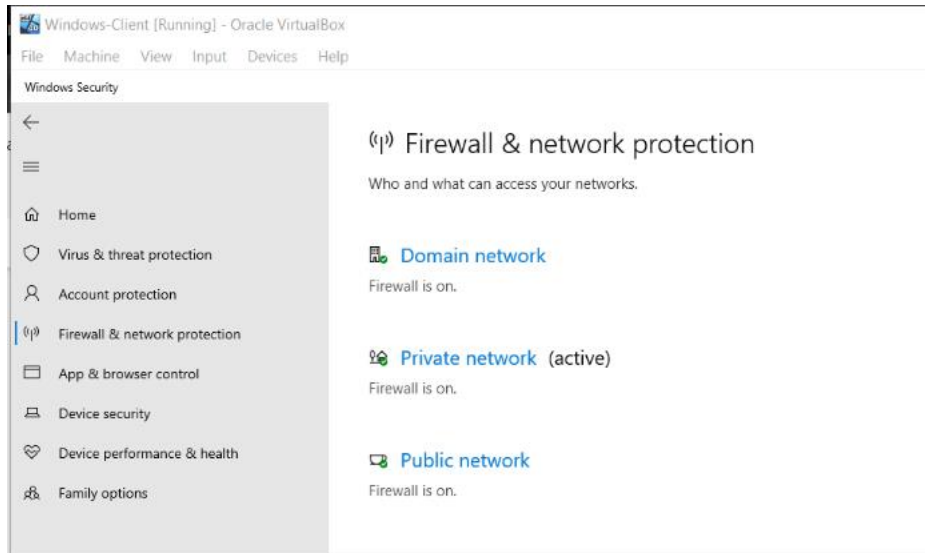
Step 2: Implement Control AC-2 (Account Management) on Windows 10

- Create or review user accounts and group memberships
- Enforce strong password policies (minimum length, complexity) via **Local Security Policy**:
- Start > secpol.msc > Account Policies > Password Policy
- Disable or remove inactive accounts



Step 3: Implement Control SI-3 (Malicious Code Protection) on Windows 10

- Ensure Windows Defender or a similar antivirus is enabled and up to date
- Run a full system scan and verify no threats are present
- Enable real-time protection and automatic updates



 **ZoneAlarm Free Antivirus**

ACTIVATE    

[< Back](#)

 **Firewall** 
Protects you from network snoopers and attacks.
Your public and private networks are protected.

 Two-way Firewall

 Application Control

 Port Protection

 [Zone Rules](#)

 [Firewall Settings](#)

 [Firewall Events](#)

 **Application Control** 
Enables you to allow or block applications from connecting to the internet

 [Application Permissions](#)

 **Anti-Bot** 
Blocks malicious communications and stops bot threats

[Send feedback](#) [Your version 4.3.283.19962 is up to date](#) [Events timeline >](#)

 **ZoneAlarm Free Antivirus**

ACTIVATE    

[< Back](#)

Events Timeline [Search:](#)

Today, 2:46:13 PM

 **Antivirus Scan**
Critical Scan (71,421 files scanned)
[DETAILS](#)
Started: Today, 2:42:06 PM
Ended: Today, 2:46:12 PM
Result: Completed
Files Scanned: 71,421

Today, 11:01:55 AM

 **AV Update Event**
AV Signatures updated
[DETAILS](#)

Yesterday, 9:39:02 PM

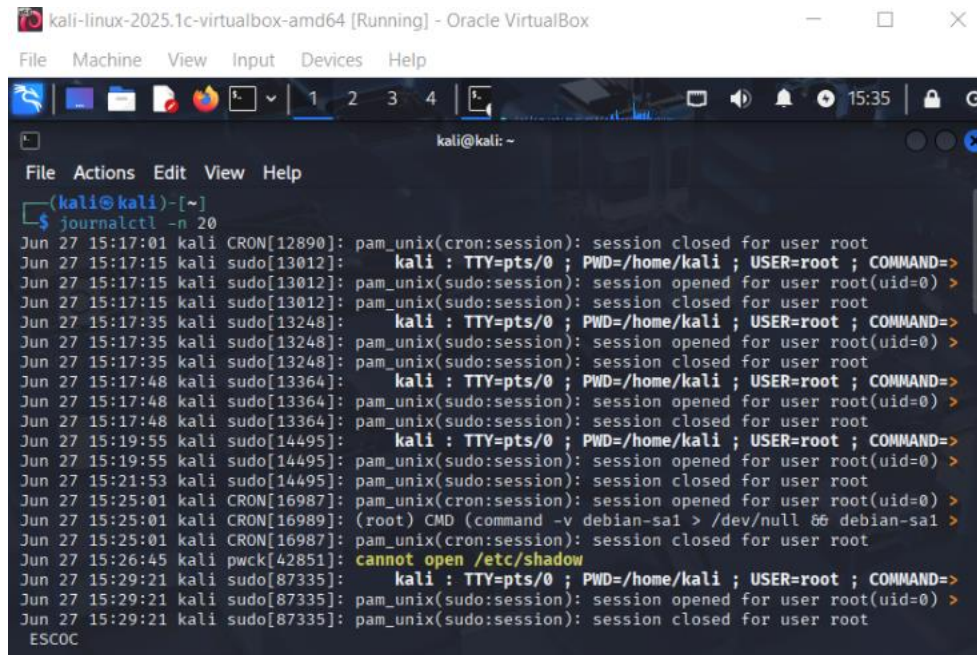
 **AV Update Event**
AV Signatures updated
[DETAILS](#)

[Send feedback](#) [Your version 4.3.283.19962 is up to date](#)

Step 4: Implement Control AU-6 (Audit Review, Analysis, and Reporting) on Kali Linux

- Configure audit logging for authentication events:
- Verify /var/log/auth.log is logging correctly

- Use ausearch or journalctl to review logs for recent login attempts
- Set up a scheduled job (cron) to archive logs weekly (provide the cron job script or command)

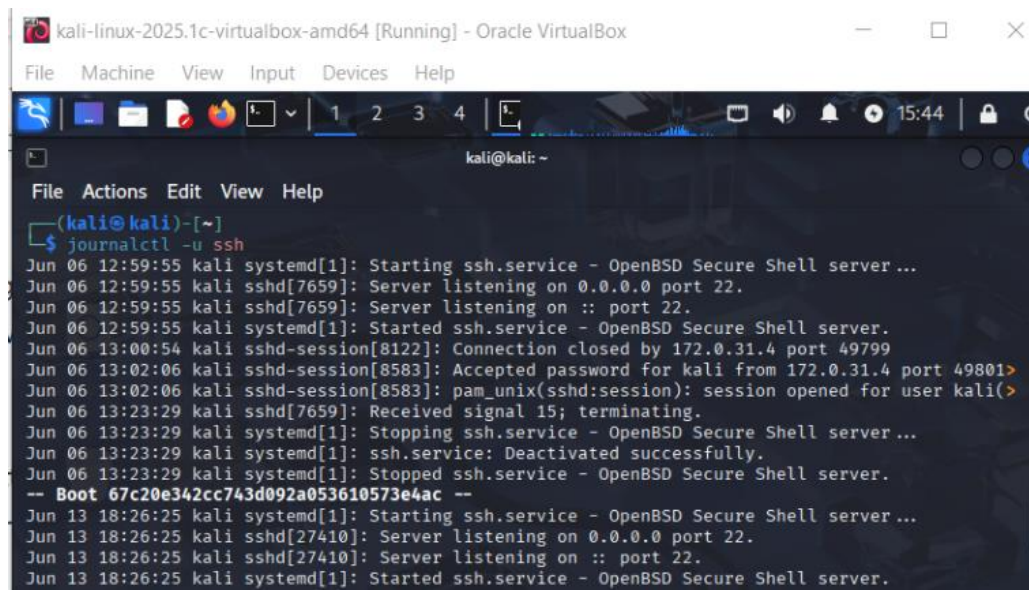


The screenshot shows a terminal window titled "kali-linux-2025.1c-virtualbox-amd64 [Running] - Oracle VirtualBox". The terminal displays the output of the command `journalctl -n 20`. The output shows a series of login attempts for the user 'root' using 'pts/0'. The attempts are timestamped from June 27, 15:17:01 to 15:29:21. The output includes messages from 'pam_unix(cron:session): session closed for user root', 'pam_unix(sudo:session): session opened for user root(uid=0)', and 'pam_unix(sudo:session): session closed for user root'. The terminal also shows a cron job running 'CMD (command -v debian-sal > /dev/null && debian-sal)' and a message 'cannot open /etc/shadow'.

```

(kali@kali)-[~]
$ journalctl -n 20
Jun 27 15:17:01 kali CRON[12890]: pam_unix(cron:session): session closed for user root
Jun 27 15:17:15 kali sudo[13012]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=>
Jun 27 15:17:15 kali sudo[13012]: pam_unix(sudo:session): session opened for user root(uid=0) >
Jun 27 15:17:15 kali sudo[13012]: pam_unix(sudo:session): session closed for user root
Jun 27 15:17:35 kali sudo[13248]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=>
Jun 27 15:17:35 kali sudo[13248]: pam_unix(sudo:session): session opened for user root(uid=0) >
Jun 27 15:17:35 kali sudo[13248]: pam_unix(sudo:session): session closed for user root
Jun 27 15:17:48 kali sudo[13364]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=>
Jun 27 15:17:48 kali sudo[13364]: pam_unix(sudo:session): session opened for user root(uid=0) >
Jun 27 15:17:48 kali sudo[13364]: pam_unix(sudo:session): session closed for user root
Jun 27 15:19:55 kali sudo[14495]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=>
Jun 27 15:19:55 kali sudo[14495]: pam_unix(sudo:session): session opened for user root(uid=0) >
Jun 27 15:21:53 kali sudo[14495]: pam_unix(sudo:session): session closed for user root
Jun 27 15:25:01 kali CRON[16987]: pam_unix(cron:session): session opened for user root(uid=0) >
Jun 27 15:25:01 kali CRON[16987]: (root) CMD (command -v debian-sal > /dev/null && debian-sal >
Jun 27 15:25:01 kali CRON[16987]: pam_unix(cron:session): session closed for user root
Jun 27 15:26:45 kali pwck[42851]: cannot open /etc/shadow
Jun 27 15:29:21 kali sudo[87335]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=>
Jun 27 15:29:21 kali sudo[87335]: pam_unix(sudo:session): session opened for user root(uid=0) >
Jun 27 15:29:21 kali sudo[87335]: pam_unix(sudo:session): session closed for user root
ESCOC

```



The screenshot shows a terminal window titled "kali-linux-2025.1c-virtualbox-amd64 [Running] - Oracle VirtualBox". The terminal displays the output of the command `journalctl -u ssh`. The output shows the SSH service starting and listening on port 22. It also shows a connection attempt from 172.0.31.4 port 49799, which was closed. A successful login attempt is shown for the user 'kali' from 172.0.31.4 port 49801. The output includes messages from 'systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...', 'sshd[7659]: Server listening on 0.0.0.0 port 22.', 'sshd-session[8122]: Connection closed by 172.0.31.4 port 49799', 'sshd-session[8583]: Accepted password for kali from 172.0.31.4 port 49801', 'pam_unix(sshd:session): session opened for user kali(>', 'sshd[7659]: Received signal 15; terminating.', 'systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...', 'ssh.service: Deactivated successfully.', 'systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.', and 'systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...'. The terminal also shows a message 'Boot 67c20e342cc743d092a053610573e4ac --'.

```

(kali@kali)-[~]
$ journalctl -u ssh
Jun 06 12:59:55 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jun 06 12:59:55 kali sshd[7659]: Server listening on 0.0.0.0 port 22.
Jun 06 12:59:55 kali sshd[7659]: Server listening on :: port 22.
Jun 06 12:59:55 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Jun 06 13:00:54 kali sshd-session[8122]: Connection closed by 172.0.31.4 port 49799
Jun 06 13:02:06 kali sshd-session[8583]: Accepted password for kali from 172.0.31.4 port 49801>
Jun 06 13:02:06 kali sshd-session[8583]: pam_unix(sshd:session): session opened for user kali(>
Jun 06 13:23:29 kali sshd[7659]: Received signal 15; terminating.
Jun 06 13:23:29 kali systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
Jun 06 13:23:29 kali systemd[1]: ssh.service: Deactivated successfully.
Jun 06 13:23:29 kali systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
-- Boot 67c20e342cc743d092a053610573e4ac --
Jun 13 18:26:25 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jun 13 18:26:25 kali sshd[27410]: Server listening on 0.0.0.0 port 22.
Jun 13 18:26:25 kali sshd[27410]: Server listening on :: port 22.
Jun 13 18:26:25 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

```

```
kali-linux-2025.1c-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
File Actions Edit View Help
(kali@kali)-[~]
$ sudo nano /usr/local/bin/archive-logs.sh
[sudo] password for kali:
(kali@kali)-[~]
$ sudo chmod +x /usr/local/bin/archive-logs.sh
(kali@kali)-[~]
$ sudo crontab -e
no crontab for root - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano          ← easiest
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny

Choose 1-3 [1]: 0 0 * * 0 /usr/local/bin/archive-logs.sh
Choose 1-3 [1]: 1
crontab: installing new crontab

(kali@kali)-[~]
$ sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
```

```
kali-linux-2025.1c-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
File Actions Edit View Help
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
0 0 * * 0 /usr/local/bin/archive-logs.sh
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command

(kali@kali)-[~]
$ grep CRON /var/log/syslog
Jun 27 17:27:22 kali cron[518]: (CRON) INFO (pidfile fd = 3)
Jun 27 17:27:22 kali cron[518]: (CRON) INFO (Running @reboot jobs)

(kali@kali)-[~]
$
```

Course: NWIT 245 - Defending the Network

Lab Number: 06

Lab Title: Digital Forensics Lab Using Autopsy: Defending the Network Approach

Professor: Silvia Vargas

Date: 12/14/2025

1. Objective

Learn how to create a disk image from a live system, then use Autopsy for forensic analysis of network-related incidents by investigating disk images, recovering evidence, and supporting network defense strategies.

2. Tools and Requirements

- Windows 10 machine with Autopsy installed
- Access to a live system (can be your own test machine) for imaging
- Basic understanding of network security and digital forensic concepts

3. Lab Steps and Execution

Step 1: Install and Launch Autopsy

- If not already installed, download and install Autopsy from <https://www.sleuthkit.org/autopsy/>
- Open Autopsy
- Create a new case: provide a case name (e.g., "Network_Defense_Case"), base directory, and investigator name

New Case Information

Steps

1. Case Information
2. Optional Information

Case Information

Case Name:

Base Directory:

Case Type: ☒ Single-User ☐ Multi-User

Case data will be stored in the following directory:

< Back **Next >** Finish Cancel Help

New Case Information

Steps

1. Case Information
2. Optional Information

Optional Information

Case

Number:

Examiner

Name:

Phone:

Email:

Notes:

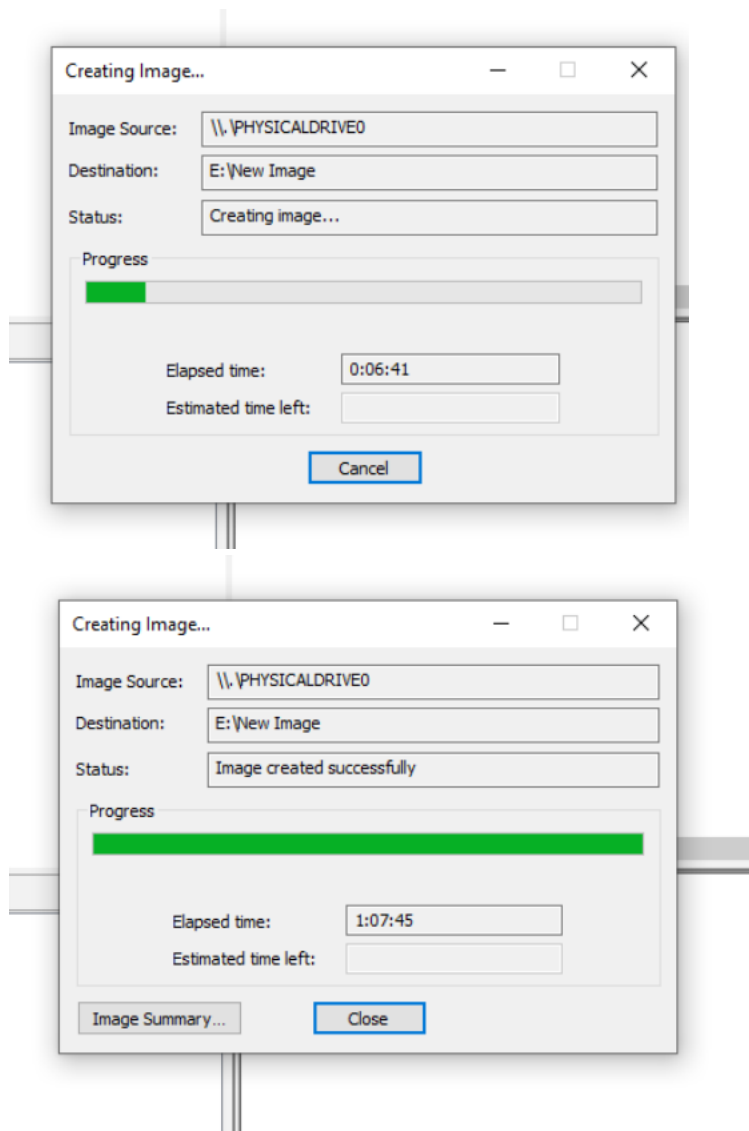
Organization

Organization analysis is being done for:

< Back Next > **Finish** Cancel Help

Step 2: Create a Simple Disk Image from a Live System

- Use **FTK Imager Lite** (free tool) or **Disk2vhd** to create a forensic image of your system or a test drive.
- Launch the tool and select **Create Disk Image**
- Choose the physical drive or partition to image
- Select **E01** or **Raw (dd)** format for output
- Save the image file to an external or separate drive



Step 3: Add Disk Image as Data Source in Autopsy

- Click **Add Data Source**
- Select the disk image file you created in Step 2
- Enable ingest modules relevant to network forensic analysis, such as:
 - File Analysis
 - Web Artifacts
 - Email Parser
 - Keyword Search
- Start the analysis

Add Data Source

×

Steps

1. Select Host

2. Select Data Source Type

3. **Select Data Source**

4. Configure Ingest

5. Add Data Source

Select Data Source

Path:

E:\New Image.E01

Browse

☐ Ignore orphan files in FAT file systems

Time zone:

(GMT-8:00) America/Los_Angeles

Sector size:

Auto Detect

Bitlocker Password (optional):

Hash Values (optional):
MD5:

SHA-1:

SHA-256:

NOTE: These values will not be validated when the data source is added.

< Back

Next >

Finish

Cancel

Help

Add Data Source

×

Steps

1. Select Host

2. Select Data Source Type

3. Select Data Source

4. Configure Ingest

5. **Add Data Source**

Add Data Source

Processing data source and adding it to a local database. File analysis will start when this finishes.

Status
Adding: Program
Files/WindowsApps/Microsoft.Office.OneNote_16001.14326.22594.0_x64__8wekyb3d8bbwe/

*This process may take some time for large data sources.

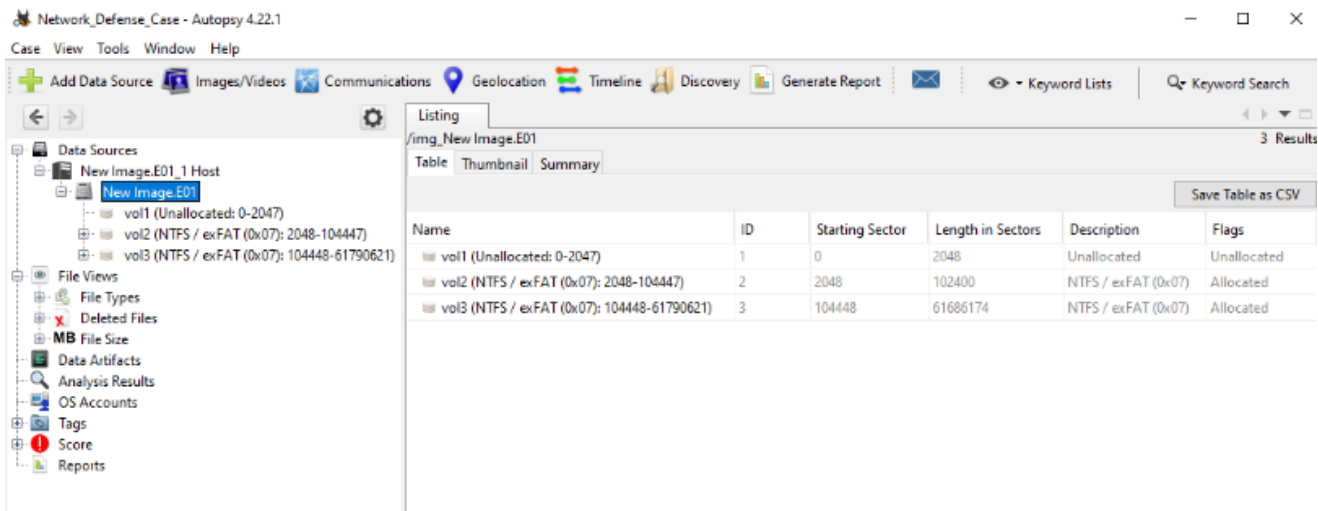
< Back

Next >

Finish

Cancel

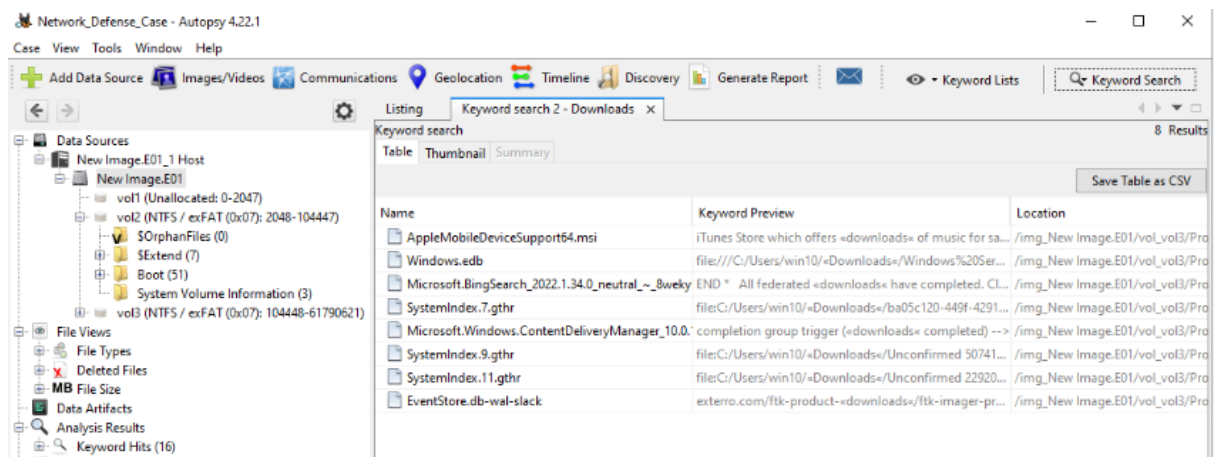
Help



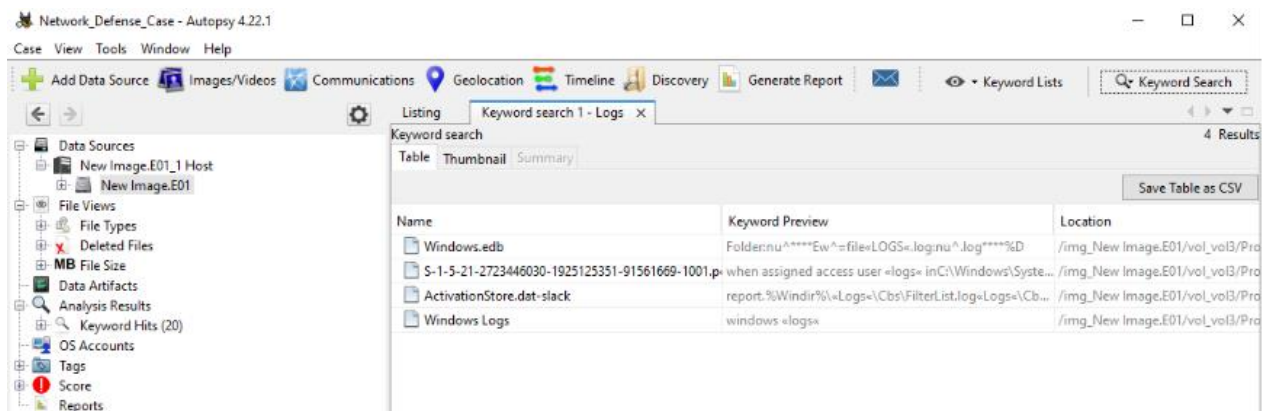
Step 4: Investigate Network-Related Artifacts

- Review key artifacts helpful in network investigations, including Keyword Searches:
 - Browser history and cache
 - Downloaded files
 - Network logs (if available)
 - Email files or chat logs
 - Recently accessed documents
- Document suspicious files or evidence of intrusion, malware, or data exfiltration

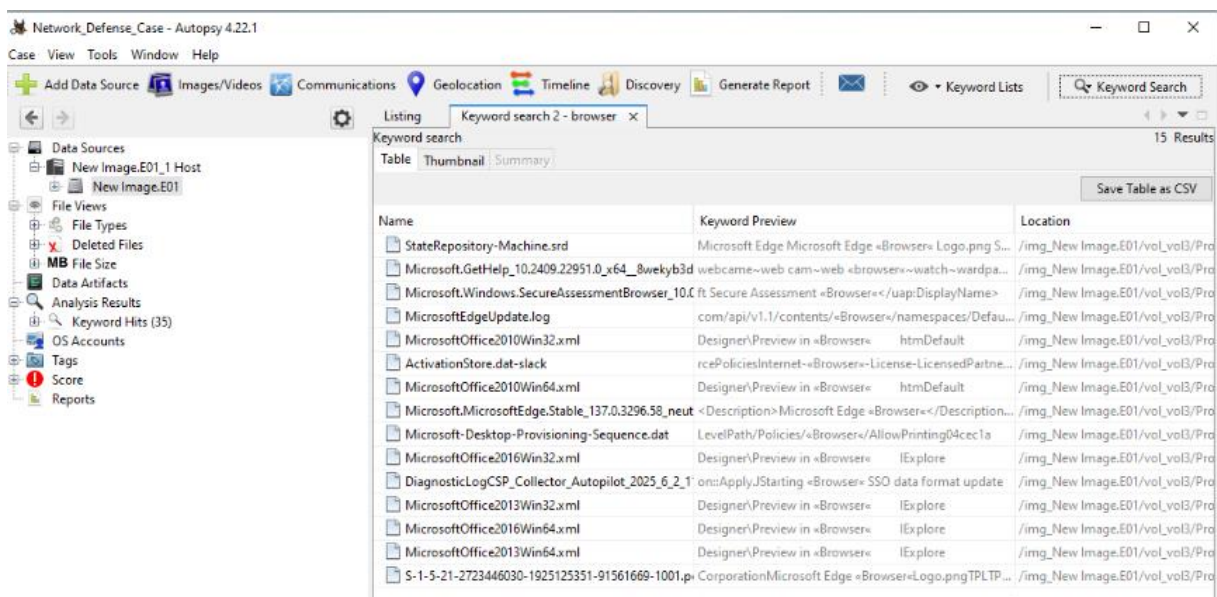
- Downloaded Files



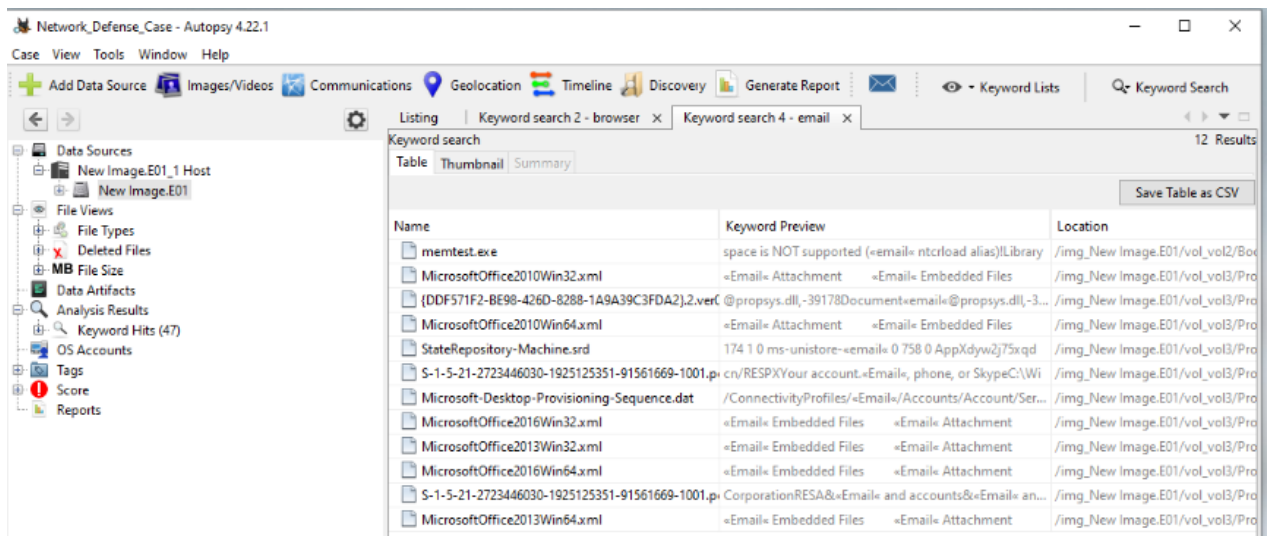
- Logs



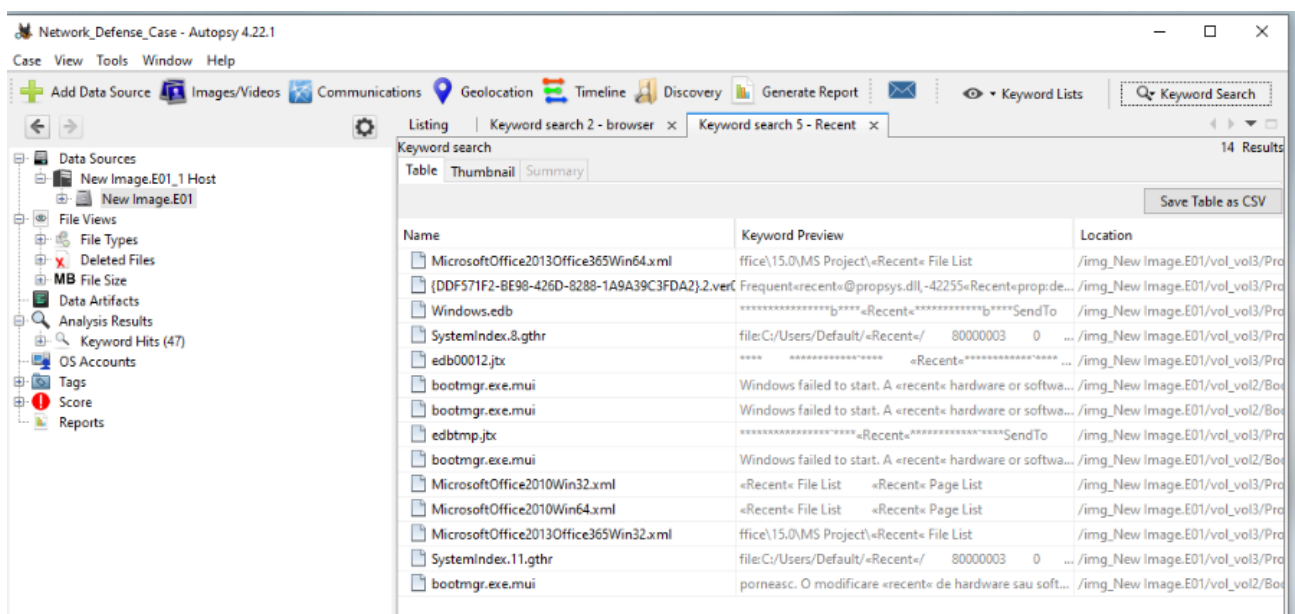
- Browser History



- Email



- Recent Files



Step 5: Generate and Export a Forensic Report

- Use Autopsy's reporting feature to generate a detailed report (HTML or PDF) summarizing findings relevant to defending the network
- Export the report and save it to your case folder
- Verify the report includes artifact summaries and search results

